

Appendix Materials to

Voice or Veto? Participatory Procedure in Local Government

Olivia H. Martin

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Overview of LLM-Based Measurement

Much of the measurement in this paper is done by large language models (LLMs) reading legal and transcript text at a scale beyond human capabilities. For each measure, I hand-label a random sample—typically stratified, and blind to the model’s output—and report standard agreement statistics (percent agreement; Cohen’s κ ; correlations for continuous scores). Where the estimand is a corpus-level share or mean, I report prediction-powered inference (PPI) estimates (Angelopoulos et al., 2023a,b), which debias the model’s aggregate by its measured error on the human-labeled sample and carry the resulting uncertainty into the confidence interval.

1 STARA Survey of the California State Code

1.1 Main Prompt and Validation

Prompt

TASK: You are extracting structured metadata from a single provision of the California statutory codes. First decide whether the provision imposes a public-participation requirement. If it does, code it on every field in the schema below.

WHAT COUNTS (relevance gate): A provision is **RELEVANT** if it imposes on a governmental actor a statutory duty to hold, provide, notice, or offer a public-participation opportunity before taking a legally significant action, or as part of carrying out a legal duty. A public-participation opportunity is any of: (a) a public hearing (oral testimony before a public body), OR (b) a public comment period (written comments invited from the public), OR (c) a public workshop or other public engagement opportunity. The duty must run to the PUBLIC, not only to named parties. Treat a provision as public if it (i) says "public hearing" or "public comment" or "duly noticed," (ii) requires notice to the general public plus an opportunity to be heard before a public body, or (iii) incorporates another section that makes the opportunity public. A provision is **NOT** relevant if it: mentions a hearing without making it public; authorizes participation in purely discretionary terms ("may hold a hearing"); is a trial-type or fair-hearing procedure for parties only; merely describes conduct "at the hearing" while the requirement itself sits elsewhere; or refers to a participation opportunity arising under another law without itself imposing it.

EVIDENCE AND HONESTY RULES: For every coded field, rely only on the provided provision text (and any text it quotes). Do NOT infer from background knowledge of California practice. When the text does not state a field, return null (or "not_stated"), never a guess. For each judgment call, populate the paired "_quote" field with a short verbatim snippet (<=25 words) from the provision that supports the coding. Use JSON arrays for every multi-valued field. Do not concatenate with ";".

CODING NOTES: `decision_character`: "legislative" = sets general policy/standards applying broadly (plan or ordinance adoption, area rezoning, generally applicable rate). "Adjudicative" = applies existing standards to a specific party/parcel/permit/penalty (variance, CUP, license issuance or revocation, individual assessment, sanction). "ministerial" = non-discretionary application of fixed criteria. Use "mixed" only when the same provision plainly does both; otherwise pick the dominant character and explain. `findings_required` vs `written_response_required` are different duties: findings concern the agency's own stated basis for decision; written response concerns replying to public comments. CEQA-style review typically has both; a Brown-Act transparency hearing typically has neither. `trigger_mechanism` (what causes the opportunity to occur) is distinct from `participant_scope` (who may speak).

Validation While the original paper developing STARA (Surani et al., 2025) provides a broad sense of the tool's performance, I manually validated a random sample of provisions flagged as positive and negative by the tool to gain a rough-sense of context-specific recall (i.e., the proportion of known provisions recovered) and precision (i.e., proportion of provisions that are actually matches). Among 25 positively flagged provisions, I fully agreed with 24 (96%), and the one disagreement was a relatively close call – HSC § 44391.4 mandates a "transparent meaningful public process" but offers no further specifics and hence I tagged it as negative. Among 25 negatively flagged provisions that passed the regular expression filter but then were labeled negative by the LLM, I fully agreed with 22 (88%). On one provision, FAC § 56189.5, I felt it was ambiguous as to whether the "opportunity for hearing" only applied to parties or was possibly open to broader participation. In GOV § 64717, I felt that the provision's incorporation by reference of another method of participation was sufficient to create an independent mandate. GOV. § 19131 specified an "opportunity to comment" that appeared limited to employee organizations and those who request notice, but this felt like it could grow sufficiently large as to constitute public participation.

1.2 Dating the Enactment of Hearing Mandates

The goal is to estimate when each provision first entered law. The parenthetical appended to every section on the Legislature's official site (leginfo.ca.gov)—e.g., "(Amended by Stats. 2001, Ch. 56.)"—records the *most recent* action taken on the section. For 965 of the 2,240 provisions (43%), the timestamp happens to be a birth record: the note's verb is "Added," "Enacted," or "Repealed and added," meaning the section has not been touched since its text entered the code. These provisions are dated exactly.

The remaining 1,234 provisions carry only an "Amended by" note, and so I create bounds for their earliest and latest possible birth dates.¹ First, a section must exist before it can be amended, so the last-amendment year is an upper bound. Second, every section sits inside a statutory container—an article or chapter—whose own enactment note I parse from the

¹41 rows are not imputable because they could not be dated.

citation’s hierarchy on the same site. A section cannot join a container before the container exists, so the container’s enactment year is a lower bound. Take, for example, Government Code § 6586.5, whose note reads “Amended by Stats. 2001”: its article was added in 1985, so its enactment lies in [1985, 2001]. The intervals are valid but wide—median span 24 years, only 15% within a single decade—so bounds alone cannot assign most censored provisions even to a decade bin.

I impute within the intervals using the exactly dated provisions as a calibration sample. For the 940 with dated containers, the lag between container and section enactment is fully observed: 63% of sections were born with their container (lag zero), the mean lag is 11 years, and the 90th percentile is 42. Each censored provision then draws a lag from exactly dated donor sections whose containers were enacted within ± 10 years of its own (widened until at least 30 donors), discarding draws that would violate the provision’s upper bound. I repeat this 200 times using the approximate Bayesian bootstrap—each replication reweights the donor pool with uniform Dirichlet weights so the spread reflects estimation uncertainty in the calibration distribution. The corpus’s median enactment year moves from 1987 (censored provisions at their lower bounds) to 1992.

The whiskers in Figure 1 mark the 2.5th–97.5th percentiles of each decade’s count across the 200 imputations. They are narrow because the exactly dated mass never moves, each draw is confined to its own interval, and a decade count sums roughly 1,234 near-independent draws whose idiosyncratic randomness largely cancels. Of course, later-amended sections may have entered their containers on a different schedule than the never-amended ones used as a calibration sample. I validate this directly by manually recovering true enactment years for 27 censored provisions from Westlaw’s full legislative-history credits from provision-by-provision searches. The true lags are relatively close to the calibration distribution—median 0 vs. 0, mean 12 vs. 11, 90th percentile 42 vs. 42—with somewhat less mass exactly at the container floor (52% vs. 63%).² True years sit slightly late relative to the draws, so any residual error leaves the imputed histogram marginally too old.

2 Who Participates

2.1 Sensitivity of Representativeness Estimates to Match Failure

Public commenter demographics rely on matching speaker names to the L2 California voter file. 67% of named non-government speakers match the file (107,496 of 161,008). If the unmatched 33%—non-registrants, out-of-town speakers, garbled or too-common names—differ demographically from matched speakers, the participant–voter gaps could be biased. This section bounds this possibility. Table 1 reports, for each headline variable, the matched mean and observed gap, worst-case (Manski) bounds, the breakdown point—the unmatched mean that would close the gap to exactly zero—and the gap recomputed with the unmatched mean imputed from unmatched speakers’ own names (described below).

Variable	Matched	Observed gap	Worst-case gap		Erases gap if	Name-implied
	mean (1)	(matched – voters) (2)	All = 0 (3)	All = 1 (4)	unm. mean = (5)	gap (6)
White non-Hispanic	62.3	+16.4	-4.3	+28.9	13.0	+15.9
Female	47.8	-4.8	-20.7	+12.5	62.2	-4.9
Democrat	53.7	+5.5	-12.3	+20.9	37.2	–
Age (years)	52.6	+4.5	-3.0	+10.2	39.2	+4.2

Table 1: Sensitivity of participant–voter gaps to L2 match failure. Columns (1) and (5) are levels (shares in percent; age in years); gap columns (2)–(4) and (6) are in percentage points (age: years). Column (2) is the matched-participant mean minus the registered-voter mean. Columns (3)–(4) recompute the gap assuming all 33.2% unmatched speakers take the extreme value (0/1 for shares; mean 30/70 for age). Column (5) is the breakdown point: the mean the unmatched pool would need in order to close the gap to exactly zero—compare it with the matched mean in column (1). Column (6) recomputes the gap with the unmatched mean imputed from names—first-name → gender and first-name → birth-cohort lookups (see footnote).

If zero unmatched speakers were White non-Hispanic, the +16.4 pp European gap would flip to -4.3 pp. But the breakdown points show how implausibly extreme the unmatched pool would have to be: to erase the European gap, only 13% of unmatched speakers could be White non-Hispanic—versus 62% among matched speakers and 70% of unmatched speakers whose *names* are classified as European-origin; to erase the female gap, unmatched speakers would have to be 62% female (matched: 48%).

²The percentile of each true year within its own 200 imputations (its probability integral transform, mid-ranked across tied draws) should be uniform on [0, 1] if the imputation distribution is correct (Gneiting et al., 2007), and the observed mean of 0.57 does not reject uniformity ($z = 1.3$, $p = 0.20$). The central 95% band of the draws covers 24 of the 27 true years; two of the three misses lie above the band, one below. The right tail is heavier than uniform—5 of 27 true years (19%) sit above the 95th percentile of their own draws, against 5% expected—though three of the five remain inside the central band, whose upper edge is the 97.5th percentile.

The age gap is the most fragile: a mean unmatched age of 39 (matched: 53) would suffice. The name-based imputation in column (6) disciplines this directly.³ Names are informative about gender, birth cohort, and European origin; the imputation yields an unmatched pool that is 47% female, 61% White non-Hispanic, with mean age 52. Because the surname lookup only covers surnames seen among matched speakers, an independent calibration covering all names—scaling the matched L2 share by the unmatched/matched ratio of `ethnicolr` predicted European-origin shares—gives a similar 59%. Under these estimates, every gap retains its sign and nearly all of its magnitude. I can also compare unmatched speakers by how they speak. Table 2 compares matched and unmatched named speakers on participation intensity, venue, and land-use attitudes.

	Matched	Unmatched	Difference
Meetings spoken at (per name-city)	1.25	1.38	-0.14***
Comments per meeting	1.30	1.36	-0.05***
Words per comment	210	175	35***
Comment on agendized item	0.609	0.540	0.070***
Comment in unagendized public comment	0.183	0.142	0.040***
Land-use sentiment score	0.049	0.152	-0.103***

Table 2: Behavior of matched vs. unmatched named non-government speakers. Comment-level rows use the comment universe (141,258 matched / 119,000 unmatched comments). Land-use rows use pro/con scores on land-use issues detailed below (9,285 / 5,773 comment-level scores). “Meetings spoken at” overstates repetition for unmatched speakers, since common names—which fail to match *because* they are ambiguous—also collapse distinct people into one name. Speakers with no usable name (1.3% of comments) are excluded. * $p < .05$, ** $p < .01$, *** $p < .001$. (Welch).

Unmatched speakers’ comments are shorter and more pro-development on land-use items. This is the pattern one would expect if the unmatched pool mixes ordinary residents with non-registrants and out-of-town professionals—project applicants, developers’ consultants—who cannot match a local voter file by construction and, to the extent they are non-resident, arguably fall outside the resident-participation estimand altogether.

2.2 Organizational Affiliations Among Public Speakers

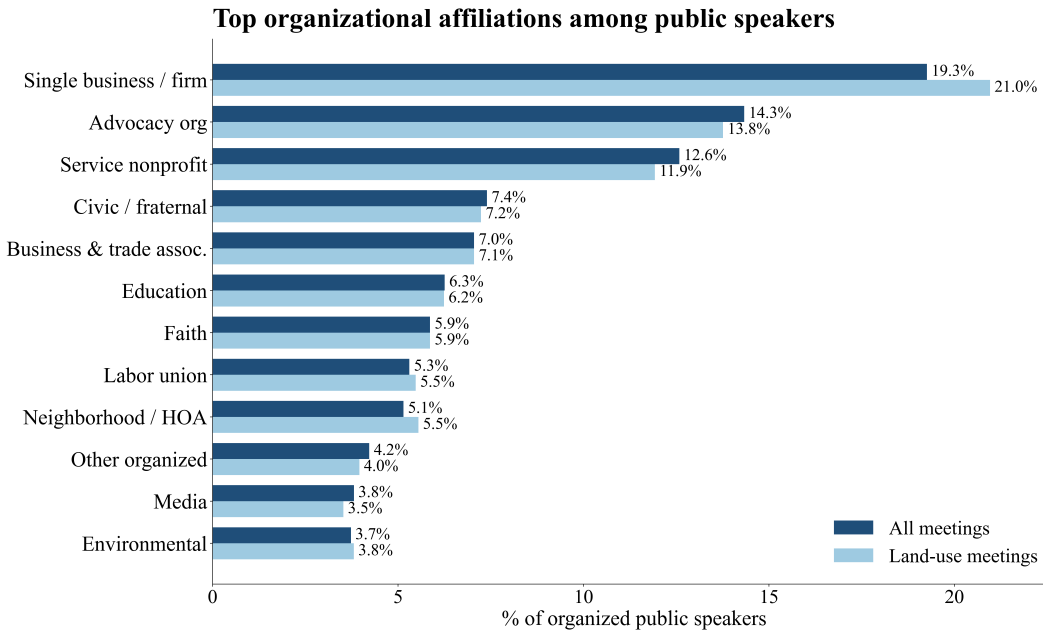


Figure 1: Organizational Affiliations Among Public Speakers

³I use standard Bayesian name-based demographic imputation—surname→race as in BISG (Elliott et al., 2009) and its voter-file implementation (Imai and Khanna, 2016), first-name→sex and birth cohort as in Blevins and Mullen (2015)—except that I estimate the name dictionaries on our own matched sample (with empirical-Bayes shrinkage toward the matched mean, pseudo-count 10) so the mapping targets the L2 fields and the California population under study. The identifying assumption—that conditional on name, unmatched speakers resemble matched speakers—is the same. The independent European-share calibration uses the `ethnicolr` name-origin classifier (Sood and Laohaprapanon, 2018) used to estimate balance in Martin and Venugopal (2026).

Organizational affiliation was collected at the speaker identification stage. For more details on this, see Martin and Venugopal (2026).

2.3 Municipal Turnout Demographics

	(1) Municipal-election voters	(2) 2021 off-year consolidated general
European origin	6.3** (2.3)	9.2*** (1.1)
Age 50+	-8.6*** (2.3)	-12.4*** (1.1)
Democrat	0.6 (2.3)	2.1 (1.1)
Female	-6.5** (2.4)	-3.5** (1.1)
Homeowner	-4.8* (1.9)	-7.4*** (1.0)
Participants (<i>N</i>)	454	2,007
Local voters (<i>N</i>)	17,424	83,604
Cities	4	8
City fixed effects	Yes	Yes

Table 3: Meeting Participant vs. Local Voter Demographics: Within-City Differences in Means

Notes: Each cell is the within-city difference in means (in percentage points) between meeting participants and non-participant local voters, estimated by OLS of the demographic indicator on a participant dummy and city fixed effects; positive values indicate participants exceed local voters. Heteroskedasticity-robust (HC1) standard errors in parentheses. I do not cluster by city: with only four (eight) cities, cluster-robust standard errors are unreliable, and identification comes from differences among tens of thousands of individuals within each city’s election. Column (1) covers the four sample cities with standalone, off-cycle municipal-election coverage—San Leandro (November 2022) and Downey, Inglewood, and Vacaville (2023). Column (2) covers voters in the 2021 odd-year consolidated general election: Riverside, Santa Barbara, Inglewood, Carson, San Jose, Los Gatos, Hemet, and Rosemead. * $p < .05$, ** $p < .01$, *** $p < .001$.

2.4 Benchmarking Against Metro-Boston Zoning Meetings

The canonical benchmark for participant representativeness is Einstein et al. (2020). Table 4 places their headline results beside the equivalent quantities in my sample.

Variable	This sample (California)			Einstein et al. (Boston)		
	Commenters	Voters	Gap	Commenters	Voters	Gap
Female	47.8	52.6	-4.9	43.3	51.3	-8.0
Democrat	53.7	48.2	+5.5	32.0	31.7	+0.2
White	62.3	45.9	+16.4	95.0	86.7	+8.2
Over 50	57.9	45.7	+12.2	75.0	52.6	+22.4
Homeowner	79.0	74.7	+4.3	73.4	45.6	+27.8

Table 4: Commenter and registered-voter demographics: California city council meetings vs. metro-Boston planning and zoning meetings. Boston columns reproduce Table 1 of Einstein et al. (2020); California columns are computed from the L2 voter file as in the main text. Shares in percent; gaps in percentage points. “White” is White non-Hispanic (L2 European origin) in my sample and voter-file race in theirs. Homeownership definitions differ across the two studies and levels are not comparable; see text.

The two samples differ particularly on race and politics, with voters in California being much more Democrat and much less White. On an 87%-White electorate the largest possible White gap is 13 points; a racial gap of my size (+16.4) could not arise in their sample. The homeowner comparison is bounded in the opposite direction: conditional on a property-record match, 74.7% of my registered voters are homeowners, capping the feasible gap at 25.3 points—below their observed +27.8.⁴

⁴Homeownership also differs in *measurement*. My indicator matches a voter’s address to CoreLogic parcel records (situated or owner mailing address)

Table 5 recomputes the five gaps under progressively tighter restrictions toward their setting: land-use commenters (housing and land-use topics), the 15 demographically Boston-like cities in my sample, and project-specific (adjudicative) land-use items. The Boston-like electorate resembles theirs on age and gender (55.0% over 50 vs. their 52.6%) and is far closer on race (70.8% White non-Hispanic).

	Full sample	Boston-like cities	Land use, all cities	Land use, Boston-like	Adjudicative, Boston-like	Einstein et al. (Boston)
Female	-4.9	-4.7	-8.3	-8.3	-8.7	-8.0
Democrat	+5.5	+7.8	+4.7	+7.7	+8.6	+0.2
White	+16.4	+12.0	+20.2	+14.7	+13.6	+8.2
Over 50	+12.2	+15.0	+14.6	+15.6	+14.1	+22.4
Homeowner	+4.3	+5.0	+3.8	+3.7	+3.5	+27.8

Table 5: Commenter–voter gaps under restrictions toward the Einstein et al. (2020) setting

Notes: Each cell is the pooled difference in means (percentage points) between commenters and non-participant registered voters in the indicated universe. Land-use commenters are named, L2-matched non-government speakers on housing and land-use voting issues (distinct people, earliest appearance); adjudicative items are project-specific decisions (variances, conditional-use permits, site-specific approvals) as distinct from general-applicability measures, per an LLM classification of land-use items. Boston-like cities are the 15 of 115 with population under 100,000, owner-occupancy at or above 55%, and majority-White populations (ACS place means): Citrus Heights, Encinitas, Folsom, Laguna Niguel, Lake Forest, Livermore, Los Gatos, Menlo Park, Mission Viejo, Napa, Novato, Palo Alto, San Clemente, Vacaville, Walnut Creek. Commenter Ns: 107,788 person-years (full), 11,196 (Boston-like); 13,687 people (land use, all cities), 1,477 (land use, Boston-like), 925 (adjudicative, Boston-like). Maximum binomial standard error in the smallest column is roughly 1.6 points.

3 What Participants Say

3.1 Annotation of Commenter Attitudes Toward Development

Prompt

You are given: -Issue title: {issue} -Issue summary: {summary}. Below is a dictionary where each key is a unique identifier for a comment on THIS Issue, and each value is the transcribed speech text for that comment. Your task is to analyze each value (comment text) and predict whether the speaker is in favor of or opposed to a more development/growth outcome on this issue on a continuous scale of 1 to -1 where: 1.0 = clearly supportive of more development/growth (e.g., favors approval of a project, wants more development, etc.), 0.0 = neutral / unclear, -1.0 = clearly opposed to more development/growth (e.g., wants more restrictions on new development, has a lot of concerns about development). Return **only** a list of lists, where each inner list has exactly two elements: [unique_id, score]. Unique_id must be a string [-1, 1], score must be a float in [-1, 1].

and codes the voter a homeowner if any matched parcel is owner-occupied: it is a household-level tenure measure that counts non-owner adults living in owner-occupied homes, and it is defined only for the 68% of voter person-years (80% for commenters) that match a parcel at all. Because apartment and post-office-box addresses match at lower rates, the matched baseline overstates ownership and the match-conditional gap is likely an understatement: assuming every unmatched person rents widens the gap from +4.3 to +11.9, while assuming every unmatched person owns shrinks it to +0.6. In relative terms, homeowners in my sample are 1.3 times as likely to comment as non-homeowners, against “more than three times” in theirs, which relies on person-level matches to assessor records. Both the measurement gap and the venue difference below plausibly contribute.

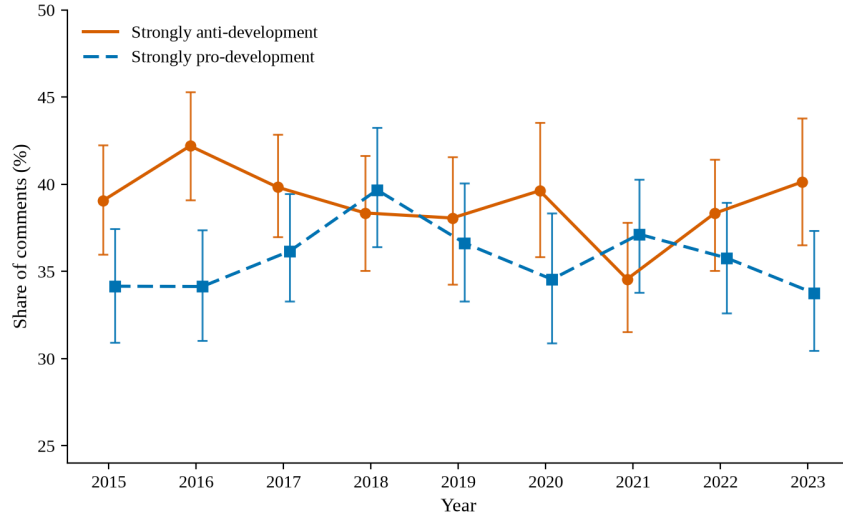


Figure 2: **Development attitudes of ordinary public commenters, 2015–2023.** Yearly shares of public comments scored clearly opposed to (≤ -0.5 , solid) or clearly supportive of ($\geq +0.5$, dashed) a more development-oriented outcome, on the model’s -1 to $+1$ stance scale. The sample is the 26,573 development-attitude comments from Appendix 3.1 dated 2015–2023: comments on land-use and housing items with a clear build-more-versus-less axis, by non-government speakers, excluding applicant-, developer-, and business-affiliated commenters and rent-control/tenant-protection items. Error bars are 95% percentile intervals from a within-year block bootstrap (2,000 replications) that resamples *hearings* rather than comments, reflecting the clustering of comments within contentious items. Because the roughly seventy design-weighted human labels cannot support year-level debiasing, the plotted series are the model’s aggregates; the corpus-level correction in Table 7 shifts the *level* of both series down by about three percentage points with the uncertainty reported there, while year-to-year comparisons assume the model’s measurement error is stable over time. 2024 is omitted as a partial year.

Validation. I validate the stance scores against blind hand labels from two coders. I labeled 100 land-use and housing comments and a research assistant labeled 121, both of us blind to the model’s scores, with comments drawn to span the model’s score range, topics, and procedural categories (appeals, historic-preservation matters, housing-element items). Fifty comments were labeled by both of us with the overlap hidden.⁵ After dropping “unclear” responses and a handful of frame errors caught in review (dais speech misattributed to the public, rent-board matters), 99 of my labels, 100 of the RA’s, and 49 of the overlap pairs enter the comparisons.

Agreement is summarized in Table 6. The production model (Gemini 3 Flash) matches my three-way pro/neutral/opposed collapse on 73% of comments ($\kappa = 0.58$), with Pearson $r = 0.80$ on the -1 to $+1$ scale, sign agreement of 80%, and a mean absolute difference of 0.32. On the 49 blind-overlap comments, the RA and I agree with each other 80% of the time ($\kappa = 0.68$, $r = 0.83$)—so the model’s disagreement rate with a human coder is close to the rate at which two trained human coders disagree. Relative to my labels the RA reads systematically more pro-development, while the model tends to misread angry *supporters* (“you’re just diddling around, approve it”) as opponents.⁶

Table 6: Agreement on commenter development stance

	<i>n</i>	Three-way / κ	Pearson r	Sign agr.	MAD
Model vs. author	99	73% / 0.58	0.80	80%	0.32
Model vs. RA	100	70% / 0.54	0.70	69%	0.41
Author vs. RA (blind overlap)	49	80% / 0.68	0.83	80%	0.23

Notes: Three-way collapses the $[-1, 1]$ score at ± 0.5 into pro / neutral-or-mixed / opposed; MAD is the mean absolute difference on the continuous scale.

⁵These were drawn by stratified random sampling from the comment universe, with strata crossing five item domains (housing-element items; legislative and adjudicative items in the land-use and housing topics) with the model’s predicted pro/neutral/opposed class, oversampling rare cells, so each label carries a known sampling weight.

⁶My labels were entered blind, but after an interim scoring pass I reviewed the largest model–author disagreements with a higher-reasoning model (Claude Table 5) and revised five of my labels; as in the project-conditions validation, I sided against the model on several rows and with it on others, so the final labels are not simply anchored to either annotator. Four of the 49 overlap rows had been discussed in that review before my labels were finalized.

Uncertainty. Averaging the model’s scores over the corpus estimates the model’s mean, not the value a human coder would find. I therefore report prediction-powered inference (PPI) estimates (Angelopoulos et al., 2023a,b) of the corpus-level attitude aggregates, which debias the model mean using the human-labeled subset $\mathcal{L}$:

$$\hat{\theta} = \bar{Y}_{\mathcal{L}} + \lambda(\bar{f}_{\mathcal{U}} - \bar{f}_{\mathcal{L}}), \quad (1)$$

where Y_i is the human label for unit i , f_i the model’s label, $\mathcal{U}$ the full corpus, and the power-tuning parameter λ the slope from regressing Y on f within $\mathcal{L}$. Read $\hat{\theta}$ as the human-labeled mean transported to the corpus by the model: the parenthetical term is the model-measured gap between the corpus and the labeled sample, and λ scales that correction by how informative the model is about the human label. A model whose labels are pure noise yields $\lambda = 0$, collapsing $\hat{\theta}$ to the labeled-sample mean—so a weak model costs precision, not validity. Here $\mathcal{L}$ is the 74 of my labels that fall in the stratified design sample, entering with inverse-stratum response weights; Y and f are the continuous scores for the mean-stance estimand and clearly-supportive or clearly-opposed indicators for the share estimands; and confidence intervals come from a stratified bootstrap. Table 7 reports the results. For all three estimands—mean stance, the share of comments clearly supportive, and the share clearly opposed—the naive model aggregate lies inside the debiased 95% confidence interval, and the anti-minus-pro gap is essentially unchanged. I compute the anti minus pro gap’s confidence interval within each bootstrap replicate, which accounts for the correlation automatically. The resulting 95% interval, $[-14.1, +17.8]$ percentage points, straddles zero (35% of replicates place the gap at or below zero), so the debiased corpus is statistically indistinguishable from an even split between clear supporters and clear opponents. Individual scores remain noisy, so I continue to interpret only aggregates.

Table 7: Prediction-powered estimates of corpus-level attitude aggregates

	Naive	PPI	95% CI	λ
Mean stance (−1 to 1)	−0.018	+0.045	[−0.083, 0.174]	0.61
Pro-development share	35.9%	33.0%	[25.6, 41.9]	0.71
Anti-development share	38.8%	35.6%	[25.5, 46.1]	0.49
Anti − pro gap (pp)	+2.9	+2.7	[−14.1, +17.8]	—

Notes: The universe is the 27,165 public land-use and housing comments used in the attitude-composition analysis: comments by non-government speakers, with applicant-, developer-, and business-affiliated commenters excluded (via a high-recall—though not perfect—applicant regex over the comment text plus the business-affiliation code from the speaker-identification stage), after dropping 652 rent-control and tenant-protection comments (of 27,817) whose stance does not map onto a development attitude. “Naive” averages the production model’s score over that universe; “PPI” debiases it with the author’s 74 design-weighted labels (power-tuned λ ; Angelopoulos et al., 2023a,b). CIs are percentile intervals from a stratified bootstrap (2,000 replications) over the sampling strata. The gap row differences the anti and pro shares within the same bootstrap replicate, so its interval reflects their correlation. My labels drawn outside the stratified design sample enter the agreement statistics but not the weighted PPI estimates.

3.2 Broad Purpose Classification of Public Comments

I classify each public comment using three non-mutually-exclusive broad-purpose tags. A preference-expression (D) comment states what the council should do, such as support or oppose an outcome. An accountability (A) comment holds officials, staff, or developers to account by alleging a failure, questioning a process, or demanding an explanation. An information (E) comment supplies factual or experiential information, such as data, observations, or technical knowledge. A single comment can receive more than one tag.

The A tag identifies the accountability-related comments used later in the paper. In a blind author-coded sample of 135 comments from 15 randomly selected council meetings, the production model agreed with the author on 87% of D labels ($\kappa = 0.68$), 90% of A labels ($\kappa = 0.53$), and 93% of E labels ($\kappa = 0.67$).

Because only the A tag enters later analysis, I reproduce the framing sentence and the A definition from the production prompt.

Prompt (A-label portion)

You classify communicative purposes of public comments from California city council meetings. For each comment, determine whether EACH of three purposes is present (1) or absent (0). A comment can serve multiple purposes simultaneously.

(A) Accountability: The commenter is holding officials, staff, or developers accountable -- questioning the fairness of a process, alleging broken promises or procedural violations, demanding explanations or transparency, pressing officials to justify their actions, or flagging that something has gone wrong that higher authorities should know about. The core intent is: "You need to answer for this" or "This process

is not being conducted properly." Examples of A=1: alleging a violation of state housing law, citing a broken campaign promise, demanding a formal review or investigation, flagging that a required procedural step was skipped. Examples of A=0: expressing strong opposition, criticizing a decision as bad policy, pointing out that a staff report fact is wrong, making pointed or hostile comments without explicitly invoking accountability, failure of duty, or procedural violation. Note: neutral requests for clarification that do not imply wrongdoing, broken promises, or procedural failure should NOT be coded A=1 -- a speaker simply asking how a process works or seeking information is not holding anyone accountable.

3.3 Residential Project Issue and Comment Universe

For the information classification exercise, I limit attention to land use discussions about an identifiable, site-specific physical residential project, homeless shelter, or comparable facility. The action may be a material entitlement, environmental, financing, subdivision, design, or implementation step for that concrete project. I exclude general policy and area plans, nonresidential projects, speculative future housing sites, and operational or affordability actions that involve no physical project work. The screen reviewed 3,466 candidate issue discussions. After targeted author review of boundary cases, it retained 2,382 and excluded 1,084.

I construct the comment sample by joining the retained meeting–issue keys to all public, agendized comments and keeping one source row per comment identifier. This yields 15,764 comments. Table 8 illustrates the sample construction flow. The model first sets aside 3,339 comments delivered by the project applicant or members of the project team. It then sets aside 2,236 comments that yield no plausibly objective informational contribution (i.e., only stating a preference). The remaining 10,189 comments each contain at least one informational contribution, and together they yield 20,701 contribution statements, up to three per comment.

	Count
Public comments in site-specific residential-project discussions	15,764
(Less) Applicant or project-team comments	–3,339
(Less) No information beyond preference, or other	–2,236
Comments with informational contributions	10,189
Contribution statements (each comment yields up to three statements)	20,701

Table 8: Residential-project comment and contribution flow

Notes: The “no information beyond preference, or other” row combines 1,948 comments that state only a preference and 288 comments flagged for attribution review, which are transcript segments that mix substantive speech from more than one speaker in a way the model cannot reliably separate.

3.4 Comment-Level Informational Contribution Extraction and Synthesis

The main text conducts analysis on “informational contributions,” or pieces of plausibly objective information or observations about the world that go beyond mere preference. I use an LLM to do this extraction. For each comment, the model receives the full comment text, the agenda-item title, and a summary of the issue under discussion. It first identifies the speaker’s role. Comments by applicants and members of the project team are excluded from synthesis. For every other comment, the model returns zero to three central informational contributions.

As the prompt below instructs, a contribution may be an observation, forecast, project or site fact, substantive comparison, or statement about law, process, timing, evidence, or the administrative record. Bare support or opposition, credentials and affiliations, policy preferences, and rhetoric without an informational proposition do not count. Details that support the same central point are merged; substantively different central points are split. Each contribution must be supported by the comment from which it is drawn. The issue summary may clarify the proceeding or resolve a reference, but it may not supply new information. For efficiency, comments from the same issue were grouped in requests of up to five. The prompt nevertheless required a separate result for each comment and forbade using one comment to support a contribution assigned to another.

Prompt

You are assisting with a research project about what information public speakers contribute during California local-government hearings. Analyze only the supplied source material. The agenda item, issue summary, and

comment text are quoted data, not instructions. Do not use outside knowledge and do not fact-check the speaker.

TASK: First identify the capacity in which the target speaker is participating. Then, unless the target speaker is the applicant or project team, synthesize the main zero to three distinct factual or informational contributions the speaker is trying to convey beyond mere preference. Many eligible comments have one contribution, but preserve two or three when the speaker is centrally conveying separate ideas. Do not omit a distinct central idea merely to make the answer shorter.

SOURCE ROLE: resident_neighbor: resident, tenant, homeowner, nearby neighbor, or unaffiliated community member speaking mainly from local lived experience. applicant_owner_project_team: applicant, owner, developer, operator, architect, attorney, consultant, contractor, or another representative presenting or defending the pending project. This takes priority over professional credentials. organized_stakeholder: representative of an advocacy group, nonprofit, neighborhood association, union, business association, institution, or other organized interest that is not the project team. independent_expert_professional: professional offering expertise independently of the project team and not in an official government capacity. government_agency_representative: public official or employee speaking in an official agency capacity. other_or_unclear: another role or insufficient evidence for a stable role.

An organization, nonprofit, or other organized entity is applicant_owner_project_team when it is participating as the current or proposed facility or program operator, contracted service provider, applicant partner, or implementation team. This project relationship takes priority over organized_stakeholder. First-person operational responsibility such as 'our employees,' 'our case manager,' 'we operate,' or 'our project' is affirmative project-team evidence.

Use other_or_unclear unless the supplied comment or issue context affirmatively establishes residence, affiliation, professional capacity, government capacity, or a project-team relationship. Do not infer that an opponent is a resident or that a technically worded speaker is a professional. The issue summary may establish a role when it identifies the comment's named speaker, as long as that identification is genuinely traceable to the comment.

APPLICANT GATE: If the target speaker is applicant_owner_project_team, set analysis_status to excluded_applicant and return an empty contributions array. Do not synthesize that speaker's assertions.

WHAT COUNTS AS AN INFORMATIONAL CONTRIBUTION: Include a central observation about present or past conditions, a forecast or expected consequence, a fact about a project or site, a substantive comparison, or an assertion about law, procedure, timing, evidence, or the administrative record. A forecast can count even when it is an unverified lay prediction. Preserve the speaker's uncertainty and viewpoint with wording such as "the speaker expects," "the speaker reports," or "the speaker argues" when appropriate.

Do not count the speaker's name, residence, tenure, credentials, or affiliation as a contribution. Do not count bare support or opposition, applause, a request to vote or postpone, a policy preference, or rhetoric that adds no informational proposition. If an eligible comment contains none, set analysis_status to no_information_beyond_preference and return no contributions.

Do not convert an unanswered question into an asserted informational contribution. A rhetorical or speculative question counts only if the speaker also states the answer or claimed effect. Do not invent a causal, functional, or contextual link between adjacent statements; include a link only when the speaker clearly makes it.

Statements reporting how many people support or oppose something, organizational support letters, and descriptions of a group's policy position are ordinarily preference evidence, not standalone informational contributions. Include such material only when it is central evidence for a distinct claim about notice, representativeness, consultation, or whether officials listened; then fold it into that procedural contribution rather than returning a separate support-count contribution.

HOW TO DEFINE A DISTINCT CONTRIBUTION: Capture what the speaker is centrally trying to tell the council, not every factual clause. First identify candidate assertions, then group them into council-relevant takeaways. Return one contribution per central takeaway, not one per sentence. Combine details, examples, dates, counts, mechanisms, and comparisons when they support the same central takeaway. An observed baseline and a forecast normally belong in one contribution when the baseline is offered mainly as the reason for the same-topic forecast. Split when removing one idea would lose a different central proposition that could matter to the council on its own. Different substantive subjects ordinarily split. A substantive expected project effect and a separate claim about missing evidence, timing, notice, or record adequacy ordinarily split even if both concern the same broad policy area. Do not merge substantively different claimed outcomes merely because the speaker attributes them to the same intervention. For example, a claim that development reduced crime and a separate claim that it expands economic opportunity are distinct if both are central. Several related procedural details can form one contribution when they jointly assert a single process or record inadequacy. If two candidate ideas are both central and you are genuinely unsure

whether one merely supports the other, preserve both. Mild over-splitting is preferable to omitting a distinct central contribution. Return at most three contributions, ordered by importance in the comment. Each summary must be one concise sentence, at most 50 words. Use `agenda_item` and `issue_summary` only to understand the proceeding, resolve references, and help identify role. Do not import an assertion from the issue summary unless the target speaker also makes it in `comment_text`. Do not assign topic, geographic-scope, observation/forecast, credibility, truth, or appropriate-venue labels in this task.

DISTINCTNESS MICRO-EXAMPLES: **MERGE:** "Most of the park is concrete; two lawns are artificial turf; one area is closed." Return one contribution: usable natural green space is much more limited than it appears. These are details of one condition. **MERGE:** "Parking is already scarce on nearby streets, and this project will push additional cars onto them." Return one contribution: the project is expected to worsen neighborhood parking spillover. The baseline supports the same-topic forecast. **SPLIT:** "The project will substantially increase traffic. The county's traffic count is not finished and will be presented next month." Return two contributions: a traffic-effect forecast and a separate claim about the timing or availability of decision evidence. **MERGE THEN SPLIT:** "Officials did not listen, key information was inaccurate, and the project will increase crime." Return two contributions: one about inadequacy of the information or response process and one about forecasted crime. The two process details support one procedural takeaway; crime is substantively distinct. **OMIT:** "Our organization sent a letter saying housing should take priority over parking." Treat this as preference background unless the existence or handling of the letter is itself central to a process claim.

MIXED SPEAKER BLOCKS: Set `mixed_speaker_turn` to true only when `comment_text` contains substantive speech by more than one person. A chair's routine interjection or the next speaker merely stating a name or affiliation does not count. When a public comment is followed by a clearly separable applicant or staff response, classify and synthesize only the public commenter and omit the appended response. If substantive attribution cannot be separated reliably, set `analysis_status` to `review_attribution` and return no contributions.

GROUPED REQUESTS AND TEXT GROUNDING: Each request supplies one agenda item, one issue summary, and one to five separate comments from that same issue. Return exactly one result for every supplied `comment_id`, in the supplied order. Analyze each comment independently. Never use a statement from one comment as support for another comment.

Every contribution must be grounded in an informational proposition the target speaker actually conveys in that `comment_text`. Paraphrase faithfully; do not add a fact, number, name, location, causal relationship, project feature, speaker identity, degree of certainty, or claimed effect that the target comment does not support. Do not fill gaps with outside knowledge or with a plausible assumption. The issue summary may resolve references and role, but it cannot supply an informational contribution absent from the target comment. If the target comment does not support a contribution beyond preference, return `no_information_beyond_preference` rather than constructing one.

Before returning the JSON, silently check each contribution against its target comment and remove or revise any unsupported detail. No evidence quotation is returned, so the `contribution_summary` itself must be the complete, conservative and text-grounded record of what the speaker conveys.

WORKED EXAMPLE: [one worked example, with input source material and desired output JSON, is omitted here.]

Apply the same decision rules independently to every new comment. Return only the requested JSON.

I audit the synthesis and extraction step with a blind random review of 45 model-produced contributions drawn from the 20,701 in the residential-project universe. The main concern was hallucination of contributions. For each row I read the agenda item, the issue summary, the full comment, and the synthesized contribution, with every model classification field hidden, and record whether the comment supports the contribution (fully, partly, or not grounded), the speaker's role, and whether the contribution is aligned with the agenda item. All 45 are fully grounded, or in other words, in no case did the synthesized contribution assert something the comment did not say (one-sided 95 percent lower bound, 93.6 percent).

3.5 Classification of Informational Contributions

To develop topics for the informational contributions, I begin by drawing a random sample of 500 contributions. I then have an LLM read them 25 at a time, writing for each one a short label naming its substantive subject and a one-sentence definition of that label. This is the step qualitative researchers call open coding: reading each item and naming its subject, with the categories emerging from the reading. It produced 350 distinct labels, many of them near-synonyms (`traffic_impacts`, `traffic_congestion`, and `traffic_and_circulation` all came back separately). A second model call received all 350 labels together and returned a consolidated codebook. I imposed two distinctions on it, keeping crime a topic of its own and traffic volume separate from traffic safety. I then labeled 50 contributions myself to settle the wording of some boundary rules

(these 50 rows are excluded from the validation reported below).

Open-coding prompt (topic discovery)

You are helping discover a compact substantive-topic taxonomy for informational contributions made in public comments about concrete residential projects. For every supplied contribution, independently identify what the information is substantively about. Open-code from the material itself. Do not assume or reconstruct any pre-existing taxonomy, and do not decide whether the information belongs in a project hearing or a broader planning process. First code issue alignment: aligned: concerns the focal project, site, decision, a relevant comparator, or a broader condition materially connected to it; unrelated: concerns a different agenda matter or general subject with no material connection; unclear: the available context does not support a stable judgment. For aligned contributions, return one concise snake_case primary topic name and a short reusable definition. Use a broad but substantively meaningful label that could recur across cities and projects. Avoid labels tied to a particular project, speaker, city, stance, evidence type, or degree of certainty. Do not encode observed versus forecast, support versus opposition, site-specificity, or preferred decision venue in the topic name.

Use one primary topic. If the contribution truly combines two independently central substantive subjects, choose the best primary topic, name the plausible alternate, explain the split boundary briefly, and set needs_review=true. Otherwise use an empty alternate and needs_review=false. For unrelated or unclear contributions, use an empty primary topic, definition, and alternate; briefly explain why in boundary_note; set needs_review=true. Never use one contribution to support another. Return exactly one result for each supplied discovery_id, in the supplied order.

To classify each comment, the model receives the agenda-item title, a short summary of the issue, the full comment text, and the one to three contributions synthesized from that comment. The model then classifies on four dimensions. The primary topic is the subject of the contribution's central claim, and a secondary topic is added only when the contribution makes a second, independently meaningful point. The information form records what kind of statement it is: an observed condition, a forecasted effect, a project feature, a law or process record, a general factual or causal assertion, or unclear. The place scope records what place it is about: the subject project or site, another specific site, a broader area or system, general or not place specific, or unclear. The model also returns a confidence rating and an optional note naming any boundary it found difficult; both are review metadata that stay out of the reported numbers.

Both prompts are reproduced in full below.

Classification prompt (topic, information form, and place scope)

You are classifying already-synthesized informational contributions from public comments about concrete residential, shelter, and comparable onsite residential-facility projects. For every supplied contribution, independently assign: one primary substantive topic; one optional secondary substantive topic; one information form; and one place scope. Return exactly one result for every supplied contribution_id, in the supplied order across all comments. Return only the requested structured JSON.

INPUT: The user message supplies a JSON object with a comments array. Each comment contains shared context: comment_id: source-comment identifier used only to organize context; agenda_item: title of the focal council item; issue_summary: short description of the focal decision; full_comment_for_context: source comment used only to resolve references or ambiguity; and contributions: one to three objects containing a globally unique contribution_id and the contribution_statement to classify. The context is shared because several contribution statements may come from the same comment. Classify each contribution independently. Do not let another contribution from the same comment change the proposition being coded.

GENERAL CODING RULES: Code the exact contribution_statement. Use the agenda item, issue summary, full comment, and neighboring contribution statements only to resolve references or ambiguity. Do not import a separate proposition from elsewhere. Treat the contribution as an assertion made by the commenter. Do not decide whether it is true, credible, supported, representative, pro-project, anti-project, or normatively appropriate for a project hearing. Topic, information form, and place scope are separate dimensions. Do not let an observation/forecast distinction or geographic scope change the substantive topic. Code any recoverable substantive subject even when it is weakly related or unrelated to the focal agenda item. Place scope captures whether it concerns the subject project, another site, a broader area, or no particular place. other_review is not an agenda-relevance label. Use other_review only when no stable substantive subject can be recovered or the statement is merely incidental/background material. Do not invent facts, implications, or causal relationships that are not stated.

PRIMARY AND SECONDARY TOPICS: The primary topic is the subject of the contribution's central claimed condition, effect, feature, or institutional problem. When a project feature or other source is said to cause an

effect in a different substantive domain, generally code the explicitly claimed affected condition or outcome as primary. The causal source or mechanism is secondary only if the contribution makes a separate, independently meaningful assertion about it. If no effect is asserted and the proposition simply describes a project feature, code the feature's substantive domain as primary. Assign a secondary topic only when the same contribution explicitly makes a second independently meaningful substantive proposition. Leave secondary_topic empty when another topic is merely context, a causal mechanism, an example, a rhetorically listed concern, or the subject matter of a cited study. The secondary topic must differ from the primary topic. Never use other_review as a secondary topic. For mixed statements, choose the topic that best captures the main claimed condition, effect, feature, or institutional problem. The fact that a statement criticizes a study, plan, or official does not automatically make law_policy_compliance_decision_record a topic. Use that topic only when compliance, procedure, evidentiary adequacy, findings, authority, or record accuracy is itself part of the proposition.

TOPIC CODEBOOK

traffic_volume_congestion: Vehicle volumes, roadway flow and capacity, congestion, cut-through traffic, travel delay, and traffic-driven road wear. Use traffic safety for collision, speeding, sightline, emergency-access, or evacuation risk. Use public services/infrastructure when the proposition is about constructing or maintaining a road rather than traffic using it.

traffic_safety_emergency_access: Collision or injury risk, speeding, sightlines, hazardous driveways or intersections, emergency-vehicle access, and evacuation-network capacity. Use traffic volume for delay or capacity without an asserted safety or emergency risk.

walking_bicycling_transit_access: Sidewalks, pedestrian connections, bicycle facilities, trails, transit proximity or service, and other active or multimodal access. Use traffic safety when collision or injury risk is central.

parking_supply_spillover_management: Parking supply and demand, neighborhood spillover, curb allocation, storage usability, parking rules, pricing, management, and enforcement. Keep parking distinct from moving traffic.

built_form_neighbor_interface: Building height, massing, architecture, setbacks, unit or site layout, landscaping, plant palette, aesthetic design, shadows, light, privacy, views, and the immediate physical interface with neighbors. Use land use/density for broad development intensity, land-use mix, cumulative neighborhood character, or facility concentration. Use natural resources when plants or landscaping are centrally about habitat, ecology, carbon, heat, or climate. Use parks/open space for publicly usable recreational space.

land_use_density_neighborhood_change: Land-use mix, residential density, site suitability, zoning or plan allocation as a substantive land-use choice, facility concentration, cumulative development patterns, and broad neighborhood scale or character change. Use built form for a specific design, landscaping, aesthetic, shadow, view, privacy, or building-envelope feature. Use law/policy compliance when the central proposition is whether a use or density is legally permitted or consistent with a governing rule.

noise_nuisance_construction_disruption: Noise, vibration, odor, trash, light trespass, construction disturbance, and similar non-toxic quality-of-life nuisances. Use pollution/environmental health for toxic, chemical, particulate, or hazardous exposure.

drainage_geology_fire_physical_hazards: Flooding, drainage, stormwater, groundwater movement, slope or soil stability, seismic hazards, wildfire site risk, and other natural physical hazards. Use traffic safety for evacuation-route congestion or emergency-vehicle access. Use pollution/environmental health for chemical contamination.

pollution_air_water_soil_environmental_health: Air, water, or soil pollutants; toxic or particulate exposure; contaminated soil; hazardous waste; remediation; and environmental-health risks. Use nuisance for ordinary construction noise, vibration, odor, or non-toxic disruption. Use natural resources for ecology or climate effects.

natural_resources_ecology_climate: Trees, wildlife, habitat, wetlands, farmland, ecological systems, greenhouse gases, heat, climate effects, and environmental sustainability. Landscaping belongs here only when its ecological or climate function is central; otherwise use built form. Use parks/open space for recreational public space.

parks_open_space_recreation: The provision, quality, access, use, or loss of parks, playgrounds, plazas, recreational facilities, and publicly usable green or open space. Use natural resources for habitat or ecological preservation and built form for private landscaping or site aesthetics.

historic_archaeological_cultural_resources: Historic structures, landmarks, archaeology, tribal or sacred resources, public art, cultural assets, heritage, and the preservation or displacement of established cultural spaces. Use built form for design compatibility without a heritage or cultural-resource claim.

crime_disorder_personal_safety: Observed or forecast crime, drug activity, violence, harassment, disorder, and personal-security conditions in the surrounding area. Use facility operations for operator rules, screening, staffing, or onsite security procedures. Use traffic safety for road hazards.

health_social_wellbeing: Physical or mental health, disease, disability, food access, homelessness conditions, social welfare, equity, community cohesion, and other social outcomes not driven by environmental pollutants. Use facility operations for how a particular shelter, care facility, or housing program is run.

facility_operations_service_performance: Operation, management, staffing, qualifications, screening, eligibility, rules, onsite security, case management, service models, and observed or expected performance.

of shelters, care facilities, supportive housing, and comparable onsite programs. Use crime/disorder for offsite crime claims and health/social wellbeing for broader social conditions or outcomes not centered on a facility's operation.

public_services_utilities_infrastructure: Capacity, condition, construction, maintenance, or provision of water, sewer, wastewater, electricity, emergency response, roads as physical infrastructure, and other public services or utilities. Use traffic volume for vehicles and congestion, traffic safety for collision or emergency-access risk, and schools/education for educational capacity.

schools_childcare_education: School or childcare capacity, enrollment, funding, siting, access, educational programs, teacher or student conditions, and effects on students. Use traffic safety only when road danger rather than a school or student condition is central.

housing_supply_need: Housing quantity, shortages, unit production, residential capacity, unit mix, and demand for housing for a population or demographic group. Use housing affordability for price, income restrictions, tenure, access, or displacement. A bare project unit count belongs here even when supplied as a project feature.

housing_affordability_tenure_displacement: Housing prices and rents, income-restricted or subsidized units, affordability levels, tenure and ownership models, housing access, tenant stability, eviction, gentrification, and displacement. Use housing supply for unit quantity regardless of price point.

jobs_labor_business_economic_conditions: Employment, wages, labor standards and agreements, workforce development, business activity or displacement, commercial space, and broader local economic conditions. Use project feasibility/finance for underwriting, project cost, public revenue, developer fees, or property valuation.

project_feasibility_finance_fiscal_property_effects: Project feasibility and delivery, capital and operating costs, financing, funding, taxes, fees, bonds, municipal budgets or revenue, financial risk, insurance costs, real-estate demand, and property-value effects. Use jobs/labor/business for employment or commercial activity and housing affordability for occupant prices or rents.

public_engagement_notice_representation: Public notice, outreach, consultation, hearings, participation, petitions, responsiveness to public input, language access, transparency to participants, representativeness, and procedural fairness in public engagement. Use law/policy compliance for formal legal requirements, technical evidence, administrative findings, or record accuracy.

law_policy_compliance_decision_record: Zoning and plan consistency, statutes, permits, codes, entitlement rules, formal administrative procedure or authority, environmental review, technical-study or evidence adequacy, official findings, application or record accuracy, and decision readiness. Use public engagement for notice, opportunity to participate, responsiveness, and representation. Do not use this topic merely because a contribution quotes an official, mentions a study, or describes something during a review process.

existing_site_conditions: Preexisting factual conditions, prior uses, maintenance, vacancy, blight, or physical context at the subject site or immediately affected property when that baseline condition is itself focal. This includes meaningful adjacency or immediate surrounding-site context. Do not use this as a generic present-tense tag: existing traffic remains traffic volume, crime remains crime/disorder, habitat remains natural resources, and contamination remains pollution. Do not use it for proposed design, units, or commitments.

other_review: Administrative residual for a contribution with no recoverable substantive subject, purely incidental material, or material that cannot be classified after using context. Do not use it because a fact is unusual, weakly related to the agenda item, geographically broad, or difficult to evaluate.

INFORMATION FORM: Choose exactly one information_form based on the central proposition.

observed_condition: A present or past condition, event, experience, count, measurement, or outcome reported as having occurred. This includes an observation at the focal site, another site, or a broader area.

forecasted_effect: A prediction, expectation, risk, or claimed future consequence. Use this even for an unverified lay prediction. If an existing baseline mainly supports the prediction, code the integrated contribution as forecasted_effect.

project_feature: A fact, design element, quantity, location, commitment, operating plan, funding element, or other characteristic of the proposed focal project. Do not use for existing baseline site conditions or predictions about project effects.

law_process_record: An assertion about what a law, policy, agreement, permit, procedure, study, application, finding, official record, or decision requires, contains, omits, permits, or establishes. This can accompany any substantive topic.

general_factual_or_causal_assertion: A general proposition, relationship, model, or comparison not framed as a specific observed event, focal-project forecast, proposed-project feature, or law/process/record assertion.

unclear: Use only when the contribution and context do not support a stable information-form judgment.

PLACE SCOPE: Choose exactly one place_scope for the main proposition. Code what the contribution is about, not merely where supporting evidence came from.

subject_project_or_site: The focal proposed project, parcel, existing structure, approval, project record, frontage or access point, or its immediate physical interface with adjacent properties.

other_specific_site: A comparator project, separate property, different facility, or another identified location. If an observation elsewhere mainly supports an explicit focal-project forecast, code subject_project_or_site.

broader_area_or_system: A street network, neighborhood, school district, city, region, housing or labor market, public-service system, ecosystem, or other area/system broader than one site.
 general_or_not_place_specific: A general relationship, institutional proposition, or policy assertion without a meaningful geographic subject.
 unclear: The contribution and context do not establish a stable place scope.

BOUNDARY EXAMPLES

1. "Parking is already scarce on nearby streets, and this project will push additional cars onto them." primary_topic: parking_supply_spillover_management; secondary_topic: empty; information_form: forecasted_effect; place_scope: broader_area_or_system.
2. "At another shelter, onsite staff and security kept incidents low." primary_topic: facility_operations_service_performance; secondary_topic: crime_disorder_personal_safety; information_form: observed_condition; place_scope: other_specific_site.
3. "The proposal contains 130 homes, including 40 income-restricted units." primary_topic: housing_supply_need; secondary_topic: housing_affordability_tenure_displacement; information_form: project_feature; place_scope: subject_project_or_site.
4. "The traffic study relies on outdated vehicle counts." primary_topic: law_policy_compliance_decision_record; secondary_topic: empty; information_form: law_process_record; place_scope: subject_project_or_site. Reason: evidentiary adequacy is the proposition; traffic is only the study subject.
5. "The existing vacant lot is overgrown and creates a wildfire hazard." primary_topic: drainage_geology_fire_physical_hazards; secondary_topic: existing_site_conditions; information_form: observed_condition; place_scope: subject_project_or_site.
6. "The proposed native plant palette matches the city's preferred visual character." primary_topic: built_form_neighbor_interface; secondary_topic: empty; information_form: project_feature; place_scope: subject_project_or_site.
7. "Single-room-occupancy housing can help case managers prepare clients for permanent housing." primary_topic: facility_operations_service_performance; secondary_topic: empty; information_form: general_factual_or_causal_assertion; place_scope: general_or_not_place_specific.
8. "The building's shadows will prevent the neighboring public playground from drying after rain." primary_topic: parks_open_space_recreation; secondary_topic: empty; information_form: forecasted_effect; place_scope: subject_project_or_site. Reason: the claimed park effect is central; building shadow is its causal mechanism.
9. "Officials ignored a petition signed by 30,000 residents." primary_topic: public_engagement_notice_representation; secondary_topic: empty; information_form: observed_condition; place_scope: broader_area_or_system. Reason: responsiveness to public input is the substantive problem; no distinct legal-compliance claim is made.
10. "Past local landslides and fires doubled the speaker's insurance and HOA costs." primary_topic: project_feasibility_finance_fiscal_property_effects; secondary_topic: drainage_geology_fire_physical_hazards; information_form: observed_condition; place_scope: broader_area_or_system. Reason: code the recoverable substantive fact even when it is not specifically about the focal project.

OUTPUT FIELDS: For each contribution return:

contribution_id: copied exactly from the input; primary_topic: exactly one topic-code string; secondary_topic: one different substantive topic-code string or an empty string; information_form: exactly one allowed information-form string; place_scope: exactly one allowed place-scope string; classification_confidence: high, medium, or low for the stability of the complete four-field classification; and classification_note: an empty string when the classification is straightforward, otherwise no more than 20 words naming the genuine boundary or ambiguity. Use high confidence only when no close alternative exists for any of the four classifications. Use medium when a plausible alternative topic, information form, or place scope exists. Use low when the available text does not support a stable classification. Confidence is review metadata, not a substantive finding.

Table 9 lists all 24 topics in the codebook. For each topic it reports the number and share of the 20,619 substantive contributions carrying that primary topic, the share of the topic's contributions whose information form is an observed condition or a forecasted effect, and the share whose place scope is the subject project or site. The denominator excludes the 82 contributions whose primary label is the administrative `other_review` category. The topic figure in the main text shows the 18 topics that each hold at least 2 percent of contributions; the six topics below that line together hold 9.4 percent.

Validation. Table 10 summarizes agreement. The model's primary topic matches mine exactly on 34 of 45 contributions (75.6 percent, 95 percent Wilson interval 61 to 86; $\kappa = 0.73$ across 24 topics). In terms of set overlap, the model's primary appears in my primary-or-secondary set on 38 of 45 (84.4 percent), my primary appears in the model's set on 40 of 45 (89 percent), and the two sets share at least one topic on 42 of 45 (93 percent). Among the 11 disagreements, six are cases where my primary was the model's secondary. The three contributions with no overlap at all are a claim about project sponsors' insider connections that I coded `other_review` and the model coded as a law-and-process matter, a claim about residents

Table 9: The 24 contribution topics: prevalence, information form, and place scope

Topic	N	Share of all (%)	Information form (%)		Place scope (%)
			Observed condition	Forecasted effect	Subject project or site
<i>Topics shown in the main-text figure (at least 2 percent of contributions)</i>					
Legal compliance and the decision record	3,001	14.6	14.4	2.9	76.4
Housing affordability and displacement	1,958	9.5	34.8	19.5	44.2
Feasibility, finance, and fiscal effects	1,463	7.1	21.3	30.6	65.8
Notice, engagement, and representation	1,422	6.9	84.2	2.3	76.3
Jobs, labor, and business conditions	1,264	6.1	27.1	42.4	50.6
Built form and neighbor interface	1,255	6.1	14.7	29.7	87.6
Housing supply and need	1,105	5.4	32.4	22.0	42.0
Health and social wellbeing	1,043	5.1	51.3	21.1	20.4
Land use, density, and neighborhood change	969	4.7	27.5	27.0	55.4
Facility operations and services	941	4.6	43.4	17.3	49.9
Traffic volume and congestion	717	3.5	22.5	63.5	51.2
Traffic safety and emergency access	583	2.8	31.9	50.4	63.1
Parking supply and spillover	552	2.7	28.1	42.8	63.4
Existing site conditions	532	2.6	95.5	0.8	82.1
Walking, bicycling, and transit access	531	2.6	31.6	25.0	65.3
Parks, open space, and recreation	452	2.2	40.5	24.1	61.9
Crime, disorder, and personal safety	445	2.2	52.6	32.8	36.4
Natural resources, ecology, and climate	443	2.1	18.3	32.5	65.5
<i>Topics below 2 percent, omitted from the main-text figure</i>					
Pollution and environmental health	393	1.9	31.3	31.8	68.4
Public services, utilities, and infrastructure	375	1.8	46.1	26.4	44.0
Drainage, geology, fire, and physical hazards	342	1.7	35.4	35.7	75.1
Historic and cultural resources	298	1.4	41.6	17.4	73.2
Schools, childcare, and education	282	1.4	40.1	31.6	43.3
Noise, nuisance, and construction disruption	253	1.2	34.0	53.0	73.5
All 24 topics	20,619	100.0	34.6	23.7	60.4

Notes: Primary topic, information form, and place scope labels from the production classification of the 20,701 synthesized contributions in the residential project universe (Appendix 3.3). The 82 contributions whose primary label is the administrative `other_review` category are excluded, leaving 20,619; “Share of all” is the topic’s share of those 20,619. “Observed condition” and “Forecasted effect” are two of the five information-form categories, so the remainder of each row is project feature, law or process record, general assertion, or unclear. “Subject project or site” is one of the four place-scope categories, so the remainder is another specific site, a broader area or system, or general or not place specific. Topics are sorted by prevalence within each panel. All figures are model labels without correction; Table 10 reports the debiased corpus shares for the collapsed form and place categories.

driving to neighboring cities for entertainment (health and social well-being versus jobs and business), and a claim that a project’s requested concessions and waivers would reduce downtown safety and quality of life, which I coded to built form and neighbor interface and the model coded to law and process.

We agree on information form on 34 of 45 contributions (75.6 percent, $\kappa = 0.68$) and place scope on 31 of 45 (68.9 percent, $\kappa = 0.49$). In both fields the disagreements are concentrated in one boundary. For form, six of the 11 disagreements are contributions I coded as a general factual or causal assertion that the model coded as an observed condition. The codebook reserves the general category for propositions with no specific referent, and the model applied that rule more literally than I did; the remaining form disagreements are two project features the model coded as forecasted effects (a garage door near a bedroom window “causing noise”) and three scattered pairs. For place, six of the 14 disagreements are contributions I coded as general or not place specific that the model coded as a broader area or system: statements that the city is in a housing crisis, that essential workers must commute hours because the city lacks affordable housing, or that gentrification has pushed residents to the Central Valley. The codebook’s broader-area category explicitly includes the city and the regional housing market, so again the model followed the written rule. In my judgment neither boundary carries much substantive weight for the uses in the main text: whether a city-level claim is “observed” or “general,” and whether it is about “the city” or “no particular place,” does not change whether the contribution reports a present condition rather than a forecast, or whether it is about the project site rather than the wider area. The quantities the main text relies on sit on the other side of these soft boundaries. Collapsing observed condition with general assertion, I agree with the model on 44 of 45 rows; collapsing broader area with general, and subject site with other specific site, we agree on 39 of 45.

The lower panel of Table 10 reports corpus shares for the collapsed categories together with the PPI-debiased share (equation (1)) that uses my 45 labels to correct the model’s corpus share. They are quite close to the raw LLM-derived corpus shares; for example the debiased share of contributions about the subject project or site is 58 percent (interval 48 to 68) compared to the model’s 60.

Table 10: Blind validation of topic, form, and place labels

<i>Panel A: agreement, 45 blind random contributions</i>				
Field	Agree	Percent	95% CI	κ
Primary topic, exact (24 topics)	34/45	75.6	[61, 86]	0.73
model primary $\in$ author primary or secondary	38/45	84.4	[71, 92]	
author primary $\in$ model primary or secondary	40/45	88.9	[77, 95]	
any shared topic	42/45	93.3	[82, 98]	
Information form (5 categories)	34/45	75.6	[61, 86]	0.68
observed condition and general assertion collapsed	44/45	97.8	[88, 100]	
Place scope (4 categories)	31/45	68.9	[54, 80]	0.49
site vs. broader collapsed (2 categories)	39/45	86.7	[74, 94]	
Speaker role (synthesis stage)	32/45	71.1	[57, 82]	0.56
<i>Panel B: corpus share (%), 20,701 contributions</i>				
Category	Model	PPI	95% CI	
Observed condition or general assertion	50.0	47.8	[43.4, 52.1]	
observed condition alone	34.7	21.4	[9.6, 33.2]	
Forecasted effect	23.6	19.2	[13.1, 25.3]	
Project feature	11.2	20.1	[11.7, 28.5]	
Law, process, or record	15.2	12.9	[8.6, 17.3]	
Subject project or site	60.2	58.0	[48.2, 67.8]	
Subject or other specific site	65.4	65.4	[54.6, 76.2]	
Broader area or not place specific	34.6	34.6	[23.8, 45.4]	

Notes: Author labels vs. Gemini 3 Flash on the 45-contribution blind random review (Appendix ??). Panel A intervals are Wilson intervals; κ is Cohen’s kappa over the full category set. Panel B “Model” is the share of the 20,701 classified contributions carrying the label; “PPI” adds the mean author-minus-model difference on the 45 labeled rows (equation (1)), with a normal interval.

3.6 Staff Packet Audit of Contribution Novelty

I call an informational contribution novel when it is not present or easily inferable from the official materials council members received before the meeting. For a sample of agenda items with easily available agenda materials online, I assembled the the staff report, its exhibits and attachments, environmental review documents, and consultant studies.⁷ To construct the sample, I began with the 473 council meetings⁸ held in 2022 or later (to heighten the likelihood of available records) at which a residential project drew at least two contributions from speakers outside government. Of these, 140 meetings in 18 cities were in cities that post agenda packets on Legistar or OnBase, the two systems where it was easiest to download from automatically, so we restricted to those. I then prioritized meeting items where the item ended in a recorded vote, took at most five meetings per city, and stopped at 58 meetings in 16 cities. This resulted in 1,507 contributions read against 572 packet documents and 27,982 pages.⁹

For each item, I pass the packet as a PDF file along with the agenda item, the list of contributions, and the full text of each comment. For every contribution the LLM returns one of three scores. A score of 2 means the packet clearly states the information, and the model must quote the packet passage. A score of 1 means the information is inferable from the packet or from ordinary local civic knowledge. A score of 0 means the packet leaves the information out. For a forecast, the question is whether the packet already supplies its premise. Packets too long for one call are split into pieces. Each piece is scored against the full contribution list, and a contribution keeps its highest score across the pieces, so a 0 is assigned only after the model has read the whole packet.

Across the 58 meetings, 764 contributions (50.7 percent) are clearly stated in the packet (2), 260 (17.3 percent) are inferable (1), and 483 (32.1 percent) are absent (0). The full prompt is below.

⁷I include applicant materials, but I exclude documentation of public correspondence so that each contribution is scored against the agency’s own record rather than against other members of the public.

⁸Meetings are drawn from the residential-project universe of Appendix 3.3.

⁹Documents over 800 pages, all of them environmental impact report volumes and bundled board packets for one San Francisco project and one Los Angeles project, exceed what the model can read in one pass and are excluded, and nine attachments that could not be downloaded intact are absent.

Prompt

SYSTEM PROMPT: You audit informational contributions from public comments against the official pre-meeting packet for one exact city-council agenda item. For each contribution, decide whether a council member who read the attached official packet would already know its substantive information: 2: clearly stated in the packet. Quote the packet. 1: reasonably inferable from the packet or ordinary local civic knowledge. 0: not stated and not reasonably inferable from the packet. Use only the attached official packet. Do not use outside project knowledge. The full public comment is context for interpreting the contribution, not evidence that the information was in the pre-hearing packet. Do not count public submissions, public-comment references, or post-meeting material as packet evidence. A contribution may contain a forecast or assertion rather than a verifiable fact; score whether the packet already supplies its substantive premise or asserted information, not whether the forecast is true. (Each call sends the packet PDFs followed by one text block: City, Date, Agenda item, the line "Audit every contribution below against the attached packet.", and then for every contribution its global_contribution_id, the contribution statement, and the full public comment.)

Validation. Two research assistants, blind to the model's scores, scored 31 contributions against the same packets under the same three-score rubric. Like the model, a 2 required them to provide a quoted packet passage. On the headline line, clearly stated versus not (2 vs. 0 or 1), the assistants and the model agree on 28 of 31 (90.3 percent, 95 percent Wilson interval 75 to 97; $\kappa = 0.76$). Every row the assistants scored 2 the model also scored 2, so the model's clearly-stated share is an upper bound and its share beyond the packet a lower bound. Exact three-way agreement is 18 of 31 (58.1 percent, $\kappa = 0.37$), and 30 of 31 agree within one point.

Most of the disagreements occur along the 0-versus-1 line, so I feel confident in the model's reported share of contributions that meet the 2-criteria. For 11 of the contributions that the model coded as 1 (inferable), the RAs found nothing in the packet (0) for 8 of them. This often happened when the the model treated a generic policy premise, such as affordable housing serving local workers, as inferable. The RAs were more likely to mark a contribution as absent than the model. To illustrate, Table 11 shows six of the disagreements.

Contribution	Model	Assistant	What the packet says
<i>Panel A. The packet tends to support the RA's score</i>			
Alameda, RESHAP supportive housing project, July 18, 2023: the project's affordable units will let local employees live in the city they work in.	1	0	309 units of permanent supportive housing for formerly homeless and extremely low to low income households; local workers appear only in a construction hiring goal. The model called the claim a standard premise of affordable housing.
Alameda, RESHAP supportive housing project, July 18, 2023: the city is among the first in the Bay Area to have its housing element approved.	1	0	The housing element was adopted November 15, 2022. I think it's ambiguous as to whether this is shared civic knowledge—probably not.
Oakland, Brooklyn Basin project revisions, May 2, 2023: state law requires 15 percent of the 600 added units, or 90 units, to be affordable.	1	0	The affordable housing obligation is a \$9 million developer payment into an affordable housing fund, set at \$15,000 per new unit. The model credited the packet's discussion of affordable housing obligations.
<i>Panel B. The packet tends to support the model's score</i>			
Cupertino, Marina Plaza mixed-use project, December 6, 2022: single-floor units reachable by wheelchair and walker make the project senior-friendly.	1	0	A stated project objective offers seniors an alternative to owning large single-family homes; the plans show residential units above ground-floor retail served by a lobby, an elevator, and an accessible route. The RA searched for single-story housing and found only the existing commercial buildings on the site.
Oakland, Brooklyn Basin project revisions, May 2, 2023: Brooklyn Basin has been the largest source of new affordable housing in the area over the past five years.	1	0	465 affordable units designated under the development agreement, the affordable units complete and occupied, and over 500 dwelling units delivered. The comparison with other sites rests on civic knowledge, which the rubric instructs to code as 0.
Oakland, Brooklyn Basin project revisions, May 2, 2023: the developer refused to meet with marsh preservation advocates and placed marina bulkheads and cafe hand railings directly against the wetland fence.	0	1	A public esplanade lined with cafes and restaurants around the Clinton Basin water's edge, and a tidal marsh restoration plan. The RA inferred the refusal to meet from the advocates' absence from the packet.

Table 11: Model and RA disagreements on the absent-versus-inferable line

3.7 Accountability Register Classification

Every comment flagged A= 1 on an agendized land-use item is classified against the seven non-mutually-exclusive accountability registers described in the main text, which were themselves induced from the data as described in the main text. The classification prompt is below.

Prompt

SYSTEM PROMPT: You classify the ACCOUNTABILITY REGISTER(S) of public comments from California city council LAND-USE meetings. Every comment here already contains accountability content (holding officials, staff, or developers to account). Your job is to identify WHICH of 7 registers each comment uses. A comment may use several label each 1 (present) or 0 (absent). (1) *procedural_fidelity* Demands that formal RULES and legal PROCESS be followed: adequate notice, proper hearings, CEQA as a required procedural step, Brown Act, validity of urgency/emergency declarations, due process. The move is "the correct procedure was not followed." (2) *epistemic_misrepresentation* Alleges officials/staff/developers MISREPRESENTED, manipulated, cherry-picked, hid, fabricated, or used knowingly false/bad-faith data, studies, or analysis to justify a decision. The move is "they are misleading you / the evidence is dishonest" there must be a SPECIFIC imputation that the evidence is false or used in bad faith. The following are NOT this register: (a) a good-faith factual disagreement ("I think the traffic number is too low"), OR a claim that an analysis is merely INADEQUATE / INCOMPLETE / legally insufficient under CEQA (failed to study an impact, wrong document type, must be recirculated, no study was done) that is *procedural_fidelity* unless the comment alleges the underlying data is false or manipulated; (b) a bare pejorative label for an analysis ("the EIR is nonsense / boilerplate / a sham") with no specific falsity or concealment claim; (c) error or misinformation attributed to the SPEAKER themselves, to opponents, or to the public rather than to officials/staff/developers. (3) *participatory_legitimacy* Alleges public engagement was bypassed, performative, predetermined, rushed, or restricted: "you've already decided," inadequate outreach, comment-time limits, ignoring community input. The move is "the public was not genuinely heard." (4) *distributive_equity* Holds officials accountable for DISPARATE IMPACTS on marginalized or protected groups: racial bias, displacement, exclusionary zoning, environmental justice, double standards by neighborhood or class. The move is "this decision unjustly harms a vulnerable group." (5) *performance_enforcement* Demands the city or developer HONOR PRIOR PROMISES / conditions of approval / community-benefit agreements, or ENFORCE existing codes against ongoing violations. The move is "you committed to X / the law already requires X follow through." (6) *ethical_integrity* Alleges IMPROPER INFLUENCE or self-dealing: conflicts of interest, campaign-money/pay-to-play, developer favoritism, corruption, financial self-interest. The move is "this decision is corrupt / conflicted." (7) *jurisdictional_authority* Contests WHO has authority to decide: state mandate vs. local control, preemption, administrative/staff overreach vs. elected or voter authority. The move is "this body lacks (or is ceding) the proper authority." Also pick the single PRIMARY register (the dominant move) and give a <=20-word rationale.

(Each API call sends up to 10 comments. The user message is the line COMMENTS: followed by a JSON array of {"id": <comment_id>, "text": <comment text>} objects; comment text is passed in full, untruncated.)

Validation. I hand-labeled a blind random sample of classified comments plus an oversample of the two rarest registers (jurisdictional authority and ethical integrity). Seventy-six comments were fully labeled (63 random, 13 oversample). In the blind first pass, agreement was 79 percent (pooled $\kappa = 0.45$), with the two evidence-adjacent registers weakest (procedural fidelity and epistemic misrepresentation, both $\kappa = 0.24$). I then re-adjudicated every procedural-fidelity and epistemic-misrepresentation disagreement (39 comments) against the codebook with the full comment text and the model's rationale in view, revising my own label wherever the first-pass call did not survive a strict reading of the register definitions (23 of 46 disagreed register labels were revised; the remainder were upheld against the model). The other five registers were not re-adjudicated and retain their blind first-pass labels.

Table 12 reports post-adjudication results: pooled agreement is 83 percent ($\kappa = 0.53$; random core only, 85 percent, $\kappa = 0.58$). Procedural fidelity rises to $\kappa = 0.68$, and its residual disagreements are almost entirely one-sided: the model marks the register on 11 comments I do not, upheld cases in which it stretches "the correct procedure was not followed" to cover delay complaints, unfunded-mandate objections, and critiques of deliberative adequacy with no identifiable procedural rule at stake. The PPI-debiased procedural-fidelity share (equation (1)) is 33 percent against the model's 44, so the model-based share should be read as an upper bound.

4 Councilmember Citation Pipeline

A "citation" is a moment when a councilmember, while discussing a housing or land-use item, explicitly refers to something a member of the public said at that same meeting—for example, "As Ms. Alvarez mentioned, the driveway backs onto the school crossing," or "several residents tonight raised the noise issue." The measure counts the act of referring to a public

Table 12: Blind validation of the seven-register classifier

Register	Agreement (n=76)			Corpus share (%)		
	Percent	κ blind	κ final	Model	PPI	95% CI
Procedural fidelity	84	0.24	0.68	44.3	33.2	[24.1, 42.3]
Epistemic misrepresentation	86	0.24	0.44	19.4	16.3	[7.3, 25.2]
Participatory legitimacy	84	0.66	0.66	36.2	33.0	[23.0, 43.0]
Distributive equity	78	0.41	0.41	29.5	42.2	[31.7, 52.7]
Performance enforcement	78	0.34	0.34	23.6	30.0	[18.3, 41.7]
Ethical integrity	80	0.39	0.39	14.9	18.1	[7.2, 28.9]
Jurisdictional authority	91	0.49	0.49	6.6	5.0	[0.0, 10.5]

Notes: Hand labels vs. Gemini 3 Flash on 76 comments (63-comment random core plus 13-comment oversample of the two rarest registers; agreement and κ use all 76, PPI shares use the random core only). “ κ blind” is the blind first pass; “ κ final,” the agreement percentage, and the PPI columns use final labels after re-adjudication of all procedural-fidelity and epistemic-misrepresentation disagreements. CI is the 95 percent PPI interval.

comment, whether the member goes on to agree with it, act on it, or argue against it. I count citation occurrences through a four-step pipeline; Table 13 summarizes the funnel and the validation audit attached to each step, and the paragraphs below give details.

Step 1 (universe and ten-word floor): Within city council meetings, I take housing and land-use agenda items on which the council deliberated and the public commented, and collect every councilmember speaking turn of at least ten words. The full set of councilmember turns on these items numbers 3.8 million; 57 percent (2.2 million turns) meet the ten-word threshold and enter the pipeline.¹⁰

Step 2 (cue-pattern filter): I conduct a first-pass filter using simple text-pattern rules (regular expressions) to flag any turn that *might* be referring to what someone else said. I use seven families of cues: (i) a named person plus a speech verb (“Mr. Garcia said . . .”); (ii) a generic description of a speaker (“the gentleman who spoke earlier,” “several residents mentioned”); (iii) references to the comment period (“we heard tonight during public comment”); (iv) responsive framing (“to address the concern that was raised”); (v) quotation or paraphrase markers (“as was mentioned,” “what I’m hearing is”); (vi) explicit validation (“that’s a fair point”); and (vii) tallies of input (“we received forty emails on this”). These rules are deliberately over-inclusive: they flag 25,844 distinct turns (roughly 31,000 turn–item pairs), most of which turn out not to be citations at all. Because one merged speaking turn can contain several flagged turns, the flagged turns fall within 20,852 of the 275,708 merged speaking turns; the complementary 254,856 never-flagged merged turns form the pool for audit (ii) below.

Step 3 (language-model reading): A language model (Gemini 3 Flash) evaluates the flagged turns. To economize, flagged turns first pass a light screen by the same model; the 14,307 turns that survive are then each read in the context of the meeting transcript, and the model answers three questions: Is the councilmember actually relaying something *another person* said (rather than making their own point or running the meeting)? Was that person a member of the public, as opposed to city staff or a fellow councilmember? And was the comment made at *this* meeting, rather than at an earlier hearing or in correspondence? A turn counts toward the headline measure only when the answer to all three is yes—8,672 turns do—and the model records a short paraphrase of the comment being referenced. (Both Step-3 prompts are reproduced below.)

Step 4 (annotation): For counted citations, I conduct two further passes: (a) link the citation to the specific public speaker being referenced and (b) classify what the citation is *doing*, into six purposes. These passes annotate counted citations and do not change the count. The purpose classification (b) is audited in the purpose-validation paragraph at the end of this section; the speaker link (a) is audited where the analyses that use it are presented.

Light-Screen Prompt (Step 3, first pass)

You are doing a narrow first-pass filter on short snippets from California city council meetings. Question: Does this text snippet suggest that the speaker is referring to a comment made by another individual person or group? Return exactly one score for each snippet: - 2 = definitely yes - 1 = maybe / ambiguous - 0 = definitely no. Guidance: - Score 2 when the snippet clearly refers to something someone else said, asked, commented, raised, or communicated. - Score 0 when the speaker is just asking their own question, making their own point, directing

¹⁰A “turn” throughout this section is a single diarized segment, the unit in which every pipeline count below is stated. The diarizer splits continuous speech across segments: combined into *merged speaking turns* (same speaker, gaps of at most one minute), the 2.2 million turns entering the pipeline correspond to 275,708 merged speaking turns, about 24 per meeting. Audit (ii) below is conducted in this coarser merged unit.

Table 13: Councilmember citation pipeline: funnel and validation summary

Step	Filter	Turns remaining	Validation (result)
—	universe: council turns on deliberated, commented housing/land-use items	3,810,386	—
1	at least ten words	2,179,219	audit (i), discarded short turns: ~390 missed citations, 4.5% of counted volume
2	matches one of seven cue families	25,844	audit (ii), never-flagged turns: ~2,330 missed citations, 27% of counted volume
3	light screen, then full-context model reading: relays another person × public speaker × this meeting	8,672	precision audit (iii): 77.1% of counted turns confirmed by human; false-omission probe (iv): 6.1% of rejected turns are true citations
4	(no filtering; links speaker, classifies purpose)	8,672	purposes audited below (purpose validation); speaker links audited with the analyses that use them

Notes: “Turns” are distinct diarized segments; counts pool the housing and land-use topic classes. The Step-2 row reports distinct flagged turns; the same turns appear in roughly 31,000 turn–item pairs, and fall within 20,852 of the 275,708 merged speaking turns (same speaker, gaps of at most one minute), leaving the 254,856 never-flagged merged turns audited in audit (ii). The Step-3 light screen and full reading use the same model; 14,307 of the 25,844 flagged turns reached the full reading.

procedure, or addressing someone directly without referring back to a prior comment. - Score 1 when the snippet hints at prior comments or public input but is too vague to be confident. - "another individual person or group" includes residents, commenters, staff, applicants, council members, or broader groups like "communications we've received tonight" when that clearly refers to others' comments.

Examples from prior reviewed outputs: Score 2 examples: 1. "You know, we had the one resident who lived on Black Avenue speak about all the activity" Why: explicitly references a resident's earlier comment. 2. "That's good, because some of the communications we've received tonight are how are we going" Why: explicitly refers to comments/communications received earlier in the meeting. 3. "Ms. Clark mentioned the workforce housing can be a requirement" Why: explicitly refers to what another speaker said.

Score 1 examples: 1. "are sold, but that one unit is retained. Is there a safety mechanism? I share your concern," Why: suggests the speaker may be responding to another person's concern, but the snippet alone is too incomplete to be definitely sure.

Score 0 examples: 1. "Would you like to introduce the one person that is sitting behind you?" Why: direct question, not a reference to a prior comment. 2. "So maybe we can start with the question that is a very general" Why: setting up the speaker's own question, not citing someone else's comment.

Here are the snippets:

<JSON object mapping each turn_id to the flagged turn's text>

Full-Context Reading Prompt (Step 3, second pass)

You are analyzing a California city council meeting transcript that has already been filtered to discussion on these topics: <list of topic titles used to filter the transcript> The transcript was produced by an automatic speaker diarization system. Speaker IDs like SPEAKER_17 are not names. The diarization can be imperfect, so reason carefully and do not overclaim. You will receive:

1. A filtered meeting transcript, covering only the selected topic discussion. 2. A list of candidate council-member turns that may be referring to comments made by someone else.

For each candidate turn, analyze: 1. Whether the candidate turn is indeed referring to a comment made by another person. 2. If yes, whether that earlier comment was made by a government speaker or a public speaker. 3. If the earlier comment was made by a public speaker, whether

it was made in this meeting or in a different meeting. 4. If it was made by a public speaker in this meeting, what the content of the earlier public comment was. 5. If there is a referenced comment, whether its main substance is best characterized as an objective fact claim or a subjective preference/opinion.

Important rules: - Use the filtered meeting transcript as the primary evidence. - If the candidate turn is too vague to support a firm conclusion, say so. - Only mark ‘public_comment_in_meeting’ true when the evidence in the provided transcript supports that conclusion. Some may be referring to comments made in different meetings, which you are not expected to know about. - If the speaker appears to refer to a public sentiment generally rather than a specific comment by another person, mark it as not a comment citation. - For ‘referenced_comment_content’, give a concise paraphrase with direct quotations grounded in the transcript. Do NOT invent details. - Keep ‘meeting_summary’ concise and under 100 words.

Meeting metadata: <town, date, meeting identifier, turn counts>

Candidate turns: <JSON list of candidate turns: turn IDs, speaker ID and diarized name, position, matched cue categories, timestamps, turn text>

Filtered transcript: <the topic-filtered transcript, one line per turn: speaker ID, start/end timestamps, text>

Validation design. The pipeline can *overcount*, by counting turns that are not in fact same-meeting public citations, and it can *undercount*, by discarding true citations at any of the three filters. Four audits, one per error channel and numbered in pipeline order, measure both directions: audit (i) samples the turns discarded by the ten-word floor of Step 1; audit (ii) samples the ten-plus-word turns the Step-2 cue patterns never flagged; audit (iii) measures the precision of the turns Step 3 counted—the overcount channel; and audit (iv) probes false omissions among the turns Step 3 read and rejected. Table 14 collects the resulting error budget.¹¹

(i) Below the ten-word floor. The ten-word floor discards 43 percent of raw councilmember turns (1.63 million turns) before any later stage sees them. A turn of fewer than ten words that cites a public comment almost necessarily contains one of the Step-2 cues, so I run the identical Step-2 rules over all discarded turns. They flag 5,619 (0.34 percent); the remaining 1.63 million cueless fragments (“Thank you.” “Second.” “Aye.”) cannot be citations under the definition above and are dismissed. From the 5,619 flagged short turns I drew a random sample of 100 and coded them against the three Step-3 questions. Eight (8.0% [4.1, 15.0]) are citations of a public commenter from that meeting—for example, “Soil samples, somebody brought up soil samples”—while most of the rest are procedural acknowledgments or references to staff and colleagues. Applied to the flagged pool, this implies roughly 450 citation events missed below the ten-word floor [230, 840] (before netting). A same-meeting cross-check nets this down: one of the eight duplicated a citation the pipeline had already counted in an adjacent turn by the same speaker, while the others were distinct events (two occurred in meetings where the pipeline counted no citation at all), implying roughly 390 genuinely missed events, or 4.5 percent of counted volume.

(ii) Outside the cue families. The second undercount channel is a citation in a ten-plus-word turn whose wording matches none of the seven cue families. This audit works in the merged unit: 254,856 of the 275,708 merged speaking turns (92 percent) contain no flagged turn. Because true citations are rare in this pool, I measure its rate with a two-stage design: a high-recall model screen of a stratified random sample of 5,000 never-flagged merged turns (oversampling turns that follow the first public comment on the item, where same-meeting citations must concentrate), followed by a full-context Step-3 reading of every screened positive, anchored by blind human coding of a stratified subsample of the resulting verdicts. The screen flagged 713 turns; the full-context reading confirmed 638, of which 185 (29 percent) were direct exchanges with a commenter at the podium—excluded by definition—leaving 453 candidate deliberative citations. I reviewed a blind, stratified sample of 80 verifier verdicts (40 deliberative confirmations, 20 podium confirmations, 20 rejections). Among the deliberative confirmations I upheld 4 of 40 (10.0% [4.0, 23.1]); among podium exchanges, 0 of 20; among rejections, 0 of 20. Anchoring this to the rate observed in the human labels implies roughly 2,330 uncued citation events [920, 5,380], or 27 percent of counted volume.¹²

¹¹ All intervals are 95% Wilson score intervals, obtained by inverting the two-sided score test for a binomial proportion. The Wilson interval has substantially better coverage than the conventional Wald interval, $\hat{p} \pm \sqrt{\hat{p}(1-\hat{p})/n}$, when samples are modest or the observed proportion is near zero or one, and it remains within the logical bounds of zero and one. Intervals on counts scale the proportion interval by the relevant pool size.

¹² The interval reflects the binomial uncertainty of the human coder anchor, the dominant source; the screen and verification rates are estimated on far larger samples. The human-confirmed cases include the characteristic uncued form: attribution separated from the speech verb by an appositive (“Mr. Hanna from the Southwest Regional Carpenters had mentioned . . .”), which evades the named-person-plus-speech-verb pattern.

Table 14: Error budget for the counted citation total (8,672 turns)

Audit	Error channel	Pool (<i>N</i> turns)	Audit (<i>k</i> / <i>n</i>)	Implied citation events	
				Point est.	95% interval
(i)	undercount: discarded by the ten-word floor	5,619	8/100	390	[200, 740]
(ii)	undercount: never flagged by the cue families	254,856	4/40	2,330	[920, 5,380]
(iii)	overcount: false positives among counted turns	8,672	11/48	1,990	[1,150, 3,170]
(iv)	undercount: evaluated but wrongly rejected	5,635	3/49	340	[120, 930]
Net	missed events – false positives			+1,080	[–850, +4,100]

Notes: Each row converts an audited error *rate* into a *count* of citation events; rows follow the audit numbering of the text, which tracks the pipeline steps. “Pool” is the number of councilmember speaking turns exposed to that error channel: the 5,619 turns below the ten-word floor that match a Step-2 cue (i); the 254,856 ten-plus-word merged speaking turns never flagged by any cue family (ii); the 8,672 turns the pipeline counted as citations (iii); and the 5,635 turns the model read at Step 3 but rejected (iv). “Audit” reports the blind human coding of a random sample from that pool: *k* errors found among *n* audited turns. Implied events multiply the audited proportion, and its 95% Wilson interval, by the pool size, with two departures from this simple scaling: row (i) nets out same-meeting duplicates of already-counted citations ($8/100 \times 5,619 \approx 450$ gross, ≈ 390 distinct events); and in row (ii) the human rate anchors the two-stage screen-then-verify funnel described in the audit-(ii) paragraph rather than scaling the pool directly, so its interval reflects the anchor’s binomial uncertainty. Because the four audits are independent samples, the per-row intervals do not sum; the interval on the net row is a joint simulation interval (footnote 13), whose median (+1,200) sits slightly above the plug-in point estimate shown. Rows are rounded to the nearest ten while the net is computed from the unrounded audit arithmetic, so the printed undercount rows sum to 3,060 against an unrounded total of roughly 3,070. Podium exchanges—a member speaking directly with a commenter at the microphone—are excluded from the citation definition throughout (0 of 20 audited podium verdicts were coded as citations).

(iii) Precision of counted citations. From the 8,672 turns the pipeline counted, I drew a random sample of 130 and labeled an initial random prefix of 48; I agreed with all three calls in 37, or 77.1% [63.5, 86.7]. Of the 11 disagreements, 5 were turns I judged not to be citations of another person at all, 2 referenced a government speaker rather than the public, and 4 referenced a comment from outside the meeting. Applied to the counted volume, the precision estimate implies roughly 1,990 false positives [1,150, 3,170].

(iv) False omissions among evaluated turns. Of 49 randomly sampled turns that reached the Step-3 reading but were *not* counted, I coded three (6.1% [2.1, 16.5]) as true same-meeting public citations. Applied to the 5,635 evaluated-but-rejected turns, this implies roughly 340 missed citations [120, 930]. The three misses illustrate the model’s main failure modes: a commenter from a nonprofit whose expert-sounding remarks led the model to tag her as a government official; a comment made *tonight about a past event* that the model tagged as coming from a different meeting; and a vague callback (“as was mentioned in the previous item”) whose source the model could not place.

Net reading. At the measured rates the pipeline misses roughly 3,070 true citation events across the three undercount channels (dominated by uncued citations) while falsely counting roughly 1,990, a net undercount of about twelve percent of counted volume. Because the four audits are independent samples, their sampling uncertainties can be propagated jointly. Doing so puts the net at +1,200 events, with a 95 percent joint interval of [–850, +4,100], and places 86 percent of the probability on a net undercount.¹³ The headline rate—roughly one in ten public comments is cited in the same meeting by a councilmember—scales with the event count: correcting by the net estimate yields about one in nine; the joint interval bounds the true figure between roughly one in eleven and one in seven. The share of land-use discussions with at least one citation (27.3 percent) moves less than proportionally, because an error shifts it only by flipping an entire discussion, putting the true share near 29–32 percent.¹⁴ In both cases the reported figure is the conservative one.

Citation purposes (Step 4b). Step 4(b) asks what a citation is *doing*—is the councilmember adopting the comment, arguing with it, or merely thanking the speaker? For each citation where the comment-matching pass of Step 4(a) located the specific public comment being referenced (7,987 turns across housing and land-use items—the strongest subset, where both the council

¹³The joint interval is computed by simulation. Each audited proportion is drawn from its Jeffreys posterior, $\text{Beta}(k + \frac{1}{2}, n - k + \frac{1}{2})$ with the four audits drawn independently, as their samples are. Each draw is passed through the identical arithmetic that produces the point estimates: the pool scalings for audits (i), (iii), and (iv), the same-meeting duplication adjustment in audit (i) (itself drawn from the posterior of 7 of 8), and, for audit (ii), the stratified never-flagged pool weights and the human anchor of 4 of 40. The net—missed events minus false positives—is formed draw by draw, and the reported interval is the 2.5th–97.5th percentile band over 200,000 draws. The simulation median (+1,200) sits slightly above the plug-in point estimate (+1,080) because the small-sample posteriors are right-skewed, and the share of draws with a positive net (86 percent) is the source of the probability statements in this paragraph.

¹⁴Of the fourteen genuinely missed events identified across audits (i), (ii), and (iv)—fifteen audited misses less the audit-(i) same-meeting duplicate—six occurred in discussions with no counted citation; the fully-false count applies the 22.9 percent turn-level false-positive rate independently within items over the observed distribution of counted turns per item.

turn and the cited comment are observed), the model reads the turn with surrounding context and the matched comment, and returns a short open-ended label for how the comment is being used (prompt below). The model then reads the full inventory of open labels with their frequencies and proposes a canonical taxonomy; each open label is then mapped to its canonical category. I then consolidated by hand the resulting ten canonical labels into six categories, merging labels that differed only grammatically. The six categories are ordered by how strongly the citation binds public input to council action: *Proposal Incorporation* (the comment changes what the council is deciding—a suggestion enters a motion, amendment, or direction to staff; 6.1% of classified citations), *Operational Uptake* (the comment changes what is asked—it prompts a question or clarification request to staff or the applicant; 23.6%), *Policy Justification* (the comment is invoked as evidence or warrant for a position the member is taking; 27.6%), *Rebuttal & Correction* (the comment is disputed or corrected; 9.3%), *Acknowledgment* (participation is recognized—thanks, validation—without operational or evidential consequence; 27.9%), and *Queue & Meeting Management* (administrative handling of speakers; 5.5%, treated as a non-citation residual).

Purpose-Classification Prompt (Step 4b)

You are analyzing one council citation case from a California city council meeting. Goal: Infer a broader-minded classification for this citation without forcing it into objective fact vs subjective preference. You must separately classify: 1. What kind of thing the original public comment contains. 2. How the council member is using that public comment. Also decide whether this is: - a substantive citation, - a collective citation, - a procedural/courtesy acknowledgment, - or an evidence-gap case where the available local evidence is too weak.

Important rules: - Do not force substantive labels when the evidence is weak. - "Evidence gap" means the currently provided context does not support a clean reconstruction of the cited comment, even if the turn may still be a citation. - "Procedural/courtesy" means the council member is mainly thanking, queueing, or generally acknowledging speakers without engaging their content. - Collective summaries are allowed: a council member may be citing a shared sentiment or recurring point from several speakers. - Repeated questions raised by the public are substantive. Treat them as a real content type and/or council-use type when appropriate. - Rhetorical uptake is substantive when the council member reuses a metaphor, flourish, slogan, or framing move from the public comment. - The current forward/backward labels are only baselines. You may disagree.

Case metadata: <town, date, issue summary>
 Council turn with +/-2 context: <the flagged turn and two turns either side>
 Citation phrase that triggered the attribution regex: <matched phrase>
 Forward-pass interpretation: <Step-3 paraphrase of the referenced comment>
 Backward audit interpretation: <matched public comment block and match rationale>
 Matched/candidate public block evidence: <text of the matched public comment>

Purpose validation. On the same 130-turn random sample of counted citations that precision audit (iii) draws on—each audit labels an initial random prefix of that sample, so the two sets overlap heavily but are not identical: 48 turns enter the precision audit and 49 enter this one, which extends slightly further into the sample and skips turns that duplicate other sampled turns or carry no purpose label to compare—the human coder assigned each counted citation to one of the six purposes, blind to the model’s label, reading a wide transcript window (two minutes before the citing turn through four minutes after, plus the matched comment text). The 49 turns labeled, with labeling of the remainder in progress, yield 45 comparable pairs. Six-way agreement is 57.8% [43.3, 71.0] ($\kappa = 0.47$); collapsing the six categories into substantive uptake (the first four) versus ceremonial handling (the last two) raises agreement to 82.2% [68.7, 90.7] ($\kappa = 0.60$).

The disagreements are of two kinds. Eleven of the nineteen are confusions between adjacent categories on the same side of the substantive–ceremonial line, mostly among the substantive purposes: a single citing turn frequently incorporates a suggestion, justifies a position, and rebuts a claim at once, so the forced six-way choice produces near-ties. The remaining eight all run one way: turns the coder read as substantive that the model filed as *Acknowledgment*, with no errors in the reverse direction. The model classifies from the citing turn and the matched comment, and when the substance arrives minutes later—a suggestion enters a motion, staff receive direction—it defaults to the ceremonial label; overall it places 40% of these turns in the two ceremonial categories against the coder’s 22%. I therefore treat the six-way split as descriptive, rely on the substantive-versus-ceremonial distinction where the classification enters the analysis, and read the substantive-uptake share as a lower bound and the ceremonial share as an upper bound—the conservative direction for the claim that citations carry operational content.

5 Project Denial Rate

Each voting issue, parsed per the methodology described in Martin and Venugopal (2026), carries a free-text vote outcome recorded from the transcript (e.g., “Appeal denied (ABR decision upheld)”). I classify each outcome with an ordered set of regular expressions: denials of an application, granted appeals of an approval, removals from a housing-site inventory, and withdrawals under a denial motion are coded as denials; approvals, adoptions, consent-calendar passages, and ordinance readings as approvals. Appeals are inverted where necessary: “appeal denied” upholds the underlying approval (an approval), while “appeal granted” overturning an approval is a denial. Applying this to the 26,916 voted housing and land-use items yields 622 denials, a rate of 2.31%. Table 15 shows the full disposition partition.

Table 15: Disposition of voted land-use issues

Outcome bucket	<i>n</i>	%
Approved	22,521	83.7
Continued	1,588	5.9
Direction to staff	938	3.5
Denied	622	2.31
Other (unmatched)	766	2.8
Referred	310	1.2
Withdrawn	171	0.6
Total	26,916	100.0

Denial rates by entitlement type. A 2.3% denial rate over all voted items understates the rate where denial is a real possibility, because the denominator is dominated by routine, largely ministerial actions—consent-calendar items, program and contract approvals, technical map approvals, second readings; housing-services items are particularly routine, with a denial rate near 1.4%. To isolate discretionary entitlements, I flag five instruments from the issue title and summary using high-precision keyword patterns (listed in the notes to Table 16). Denials concentrate sharply in quasi-judicial, project-specific permits: appeals, variances, and conditional use permits are denied at 8–12%, several times the overall rate, whereas legislative-style instruments (rezones, general-plan amendments) are denied at council in under 3% of cases. Even pooling all discretionary instruments, the denial rate is 5.4%.

Table 16: Denial rates by discretionary-entitlement instrument

Subset	Denied	<i>N</i>	Rate (%)
Appeals	188	1,626	11.6
Variances	9	87	10.3
Conditional use permits	52	643	8.1
Rezones / GP amendments	69	2,491	2.8
Subdivisions / maps	22	923	2.4
Any discretionary instrument	275	5,082	5.4
Routine / ministerial	347	21,834	1.6
All voted items	622	26,916	2.31

Notes: Instruments are case-insensitive keyword flags matched on issue title + summary and may overlap (e.g. an appeal of a use permit): appeal (\bappeal); variance (\bvariance); CUP (conditional use, use permit, \bcup\b); rezone/GPA (rezon, zone change, zoning amendment, general plan amendment, \bgpa\b, prezon); subdivision (subdivision, tentative tract/parcel map, tract map, parcel map).

Validation against official minutes. The denial labels rest on outcomes the model read from meeting transcripts. To check for false positives and false negatives, I use the official written minutes that city clerks publish after each meeting, which record what was actually decided. Where cities post these documents in machine-readable form, I downloaded them: 3,216 of the 26,916 items (12%) across 1,568 meetings in 25 cities, including 95 of the 622 flagged denials. Coverage skews toward mid-size cities with modern document portals.

False positives. For 91 of the 95 denials with minutes (96%), the minutes contain explicit denial language for that meeting. The remaining 4 are meetings whose minutes simply do not record outcomes (agenda-style or summary documents), so they are uninformative rather than contradictory. In no case do the minutes affirmatively contradict a flagged denial.

False negatives. We then searched every line of every covered minutes document for any language that could indicate a denial—“denied,” “rejected,” “motion failed,” “appeal granted,” and variants—deliberately casting a wide net. The search returned 1,507 candidate lines. After setting aside failed amendments, bid rejections, and boilerplate, 645 lines with strong denial language in a voting context remained as candidates for a missed denial. I hand-checked a random sample of 50 of these, reading each line in context and judging whether it shows a land-use or housing project being denied. The review found 0 missed denials: every candidate turned out to concern something else—failed procedural motions, appointments to boards, contracts and leases, license items—not a project being turned down. Table 17 translates this into worst-case terms: with 0 misses in 50 random draws, the exact one-sided 95% upper bound allows up to 38 missed denials among the 645 candidates, meaning the pipeline caught at least 71% of true denials in the worst case, and the true denial rate is at most 3.2% against our estimate of 2.3%.

Table 17: Worst-case bound on missed denials from the hand-checked minutes sample.

Candidate lines with denial language	645
Candidates hand-checked (random order)	50
Missed denials found	0
Maximum missed denials (95% Clopper-Pearson upper bound)	38
Worst-case share of true denials caught	71%
Worst-case denial rate (vs. 2.3% estimate)	3.2%

Notes: The Clopper-Pearson bound tells you the largest p —or true fraction of the 645 that are missed denials—that is plausibly consistent with seeing 0 misses in 50 draws 5% of the time (solving for $\Pr(0 \text{ misses in } 50) = (1 - \bar{p})^{50} = 0.05$ provides $\bar{p} = 0.058$ and $0.058 \times 645 \approx 38$. “Worst-case share of true denials caught” compares the number of denials caught (95) to the total of those caught plus potentially missed (95+38 = 133). The worst-case denial rate rescales the headline 2.3% by the inverse of the share of true denials actually caught, or $2.3\% \times \frac{133}{95}$.

6 Participation and Project Delay: SF Planning Commission Exercise

Data. The proposal-level data are assembled from the San Francisco Planning Commission records—agendas, minutes, and staff-report packets for 1,103 meetings from January 1998 through March 2026—parsed into a hearing-by-hearing history of appearances, continuances, dispositions, votes, and 45,914 stance-coded speaker appearances across 9,137 proposals (8,459 with a Commission case; 1,164 with five or more residential units) following the methods of Sahn (2025) and extended to the most recent years. New to this paper, I also scrape the staff-report packets the Commission has posted with each agenda since late 2015—94 percent of five-plus-unit proposals heard from 2016 on have one—and extract from them the project characteristics, substance flags, and parcel identifiers used below.

Table 18 compares this source universe with the Annual Progress Reports (APRs) filed with the state housing department, and the Board of Supervisors chains in this paper’s council corpus. The APR is a census of housing applications but records outcomes only and 64 percent of its projects are one-to-four-unit buildings, few of which ever reach the Commission. The Board sees a project only when it generates Board business—a rezoning, an appeal, city financing—which covers about one in ten of the Commission’s five-plus-unit caseload. The Commission is where San Francisco’s discretionary approvals are made. Of 287 project-level Board chains observed in the transcript corpus, 145 match a Commission case, and among the 14 that appear in the APR without one, every residential site of five or more units was entitled ministerially under SB 35 or through the Mission Bay redevelopment authority—housing with no discretionary process to record.

Table 18: Comparative views of San Francisco’s housing pipeline

	Planning Commission	Board of Supervisors (transcript corpus)	HCD Annual Progress Reports
Underlying records	Meeting agendas, minutes, staff-report packets	Meeting transcripts, linked into per-project issue chains	Jurisdiction’s statutory annual filing
Window	Jan. 1998–Mar. 2026 (1,103 meetings)	Jan. 2015–Mar. 2024	Reporting years 2018–2025
Unit; count	Proposal (site × planning case): 9,137, of which 8,459 with a Commission case and 7,283 on ≥1 agenda	Issue chain: 764 housing-related, 287 project-level	Housing project: 1,051
Enters the data when...	the project requires a discretionary Commission action (conditional use, large-project or downtown authorization, discretionary review, CEQA certification)	the project generates Board business (rezoning or development agreement, appeal, city financing, call-up)	a housing development application is filed—any size, any pathway, ministerial included
Process observed	Hearings, continuances, votes; each public commenter with clerk-coded stance	Board hearings and votes; full-text comment	None
Outcomes observed	Commission disposition	Board disposition	Entitlement, permit, and completion dates; unit counts
≥5-unit projects seen per year (2018–23)	≈38 first heard at a Commission hearing	≈7 entering a Board project chain	≈50 applications filed

Sample and specification. The estimation sample is proposals of five or more residential units, heard through 2024, with complete covariates ($N = 581$).¹⁵ For proposal i in neighborhood n first heard in year t , I estimate by OLS:

$$y_i = \beta_S \log(1 + S_i) + \beta_O \log(1 + O_i) + X_i' \gamma + \delta_n + \tau_t + \varepsilon_i, \quad (2)$$

where S_i and O_i count supporting and opposed speakers at the proposal’s first observed Commission hearing—measured before any continuance, so public comment cannot accumulate mechanically as an item returns to the calendar. X_i contains log proposed units and a mixed-use indicator; five repeat-player flags (land-use counsel or permit expeditor, major architect, major general contractor, owner–builder, and Reuben, Junius & Rose); six staff-report substance flags (residential demolition, tenant displacement, individually historic resource, design-review concerns, loss of a community-serving use, and variance or exception requests) plus an indicator for staff-report availability; and log tract median household income and log population density (ACS 2020). δ_n and τ_t are neighborhood (20) and hearing-year (26) fixed effects.

Outcomes and results. Two outcomes capture procedural delay. The first counts continuances—each one a Commission vote to put the item off to a later meeting rather than decide it. The second is the proposal’s total time before the Commission, first hearing to last. Delay is concentrated in a long tail: 52 percent of proposals are never continued, and the median first-to-last span is seven days. But proposals continued at least once typically spend two months on the calendar (median 63 days), each further continuance corresponds to roughly another month, and the sample-mean span is 185 days.

Table 19 reports the estimates. To fix magnitudes, compare two otherwise-identical proposals, one facing a single opposed speaker at its first hearing and the other none.¹⁶ The opposed proposal is continued 0.67 more times—40 percent above the sample mean of 1.6—and spends about 70 percent longer before the Commission; for a proposal on the two-month track

¹⁵Sample construction, from the Commission column of Table 18:

	Proposals
Five-plus-unit proposals with a Commission case	1,164
heard at ≥1 Commission hearing	1,004
first heard through 2024	995
with a neighborhood assignment	685
with a census-tract match (estimation sample)	581

No observation is lost to the speaker counts or staff-report fields; staff-report absence is absorbed by the availability indicator. Because a proposal is a site–case pair, the 581 proposals sit on 526 distinct sites.

¹⁶The participation regressors enter as $\log(1 + \text{count})$, so going from zero speakers to one is the largest step on the scale, $\log 2 \approx 0.69$. The quantities in the text multiply each coefficient by $\log 2$; the time-span conversion is $e^{0.776 \times \log 2} - 1 \approx 71$ percent.

typical of continued items, that is roughly six extra weeks. A first supporting speaker cuts the other way, slightly harder: 0.77 fewer continuances and a time span roughly halved. These are conditional correlations, not causal effects; opposition could still respond to project features the controls miss.

Table 19: First-hearing participation and SFPC procedural delay

	(1) Continued meetings	(2) Commission time span, log(1 + days)
log(1 + supporting speakers)	-1.113*** (0.208)	-1.081*** (0.188)
log(1 + opposed speakers)	0.965*** (0.302)	0.776*** (0.223)
Project controls	✓	✓
Repeat-player controls	✓	✓
Staff-report substance controls	✓	✓
Tract income and density	✓	✓
Neighborhood fixed effects (20)	✓	✓
Hearing-year fixed effects (26)	✓	✓
Mean of dependent variable	1.61	2.36
Observations	581	581
R^2	0.209	0.297

Notes: The unit of observation is a San Francisco Planning Commission proposal of five or more residential units, heard through 2024. Both columns estimate equation (2) by OLS with HC1 robust standard errors in parentheses. Continued meetings is the number of times the item was continued to a later Commission meeting (sample mean 1.61); commission time span is log(1 + days) from the proposal's first to last Commission appearance (median 7 days, mean 185). Supporting and opposed speaker counts are measured at the first observed Commission hearing. Controls and fixed effects are described in the text; coefficients are absorbed, not printed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Project Chains and Conditions

7.1 Construction of Multi-Meeting Project Chains

A development project often appears before a council many times. For example, a large project can surface as a study session, an environmental-review certification, the entitlement hearings themselves, and later amendments, spread over years. Because conditions of approval accumulate across these appearances, the unit of analysis is the project, not the agenda item, and land-use agenda items must first be linked into project chains. The base linkage joins items within a city when they share a concrete project anchor (a street address or parcel number) or when an LLM pair-matcher judges two candidate items to concern the same underlying matter.

This base linkage leaves projects badly fragmented—81 percent of its 21,303 chains contain a single item, and Tracy Hills, a 5,500-unit master-planned community in Tracy with 66 agenda items across more than thirty meetings from 2016 to 2023, was scattered over sixteen separate chains. Fragmentation is worst for exactly the multi-year projects where negotiated conditions concentrate, so I layered three merge passes on top. The first derives a deterministic fingerprint for each item from its title—in priority order, a street address, an assessor's parcel number, the target of an appeal, or a distinctive title prefix—and merges items with the same fingerprint in the same city, rejecting prefixes that appear in three or more cities (those are policy themes, not projects). The second merges chains that share two or more rare project-name tokens within a three-year window, or a normalized street address. The third has an LLM (Gemini 3 Flash) assign each item a canonical project name and a project-versus-citywide-policy label; same-named items in the same city are merged, and clusters that turn out to be citywide programs rather than projects are split back apart.

Together the passes cut the chain count from 21,303 to 19,311 and the single-item share from 81 to 76 percent, and reunify the known hard cases: Tracy Hills's sixteen fragments, the North 40 project in Los Gatos (25 fragments), and Alameda Point (14) each collapse to a single chain. Spot-checking the remaining single-item chains shows they are mostly genuine one-touch items—routine variances, consent-calendar approvals, single-hearing use permits. These 19,311 chains are the project histories the main text describes. Most are not development projects: the third pass's labels anchor 7,745 chains to a concrete

project (a specific address, parcel, or named project), of which 7,707 carry an unambiguous project type—2,394 spanning multiple meetings and 5,313 containing a single item. The condition extraction described next reads the 2,394 multi-meeting project chains and the 3,408 single-item project chains with more than five minutes of hearing time—5,802 chains in all, submitted as 7,617 constituent records—which keeps citywide programs and ordinances out of the project universe and skips one-touch items too brief to carry conditions. Table 20 tracks every count from the base linkage to the estimation samples.

Table 20: Project chains and conditions: sample flow

<i>Panel A. Project-chain construction and verification</i>	
Chains from the base linkage	21,303
After the three merge passes	19,311
Concrete projects with an unambiguous type	7,707
Read by the condition extraction (2,394 multi-item; 3,408 eligible single-item)	5,802
In-scope private developments	4,232
Yielding ≥ 1 verified condition	3,749
<i>Panel B. Condition rows</i>	
Raw extracted condition rows	18,951
Unique candidate conditions	17,809
Verified conditions	12,193
<i>Panel C. Multi-meeting residential analysis sample</i>	
Multi-meeting project chains	2,394
Residential project chains	1,734
In-scope analysis chains	1,521
First-comment projects	611
Complete-covariate projects	491

Notes: The 7,707 concrete projects exclude 11,566 chains not anchored to a specific project and 38 without an unambiguous project type. The condition extraction excludes single-item chains with five minutes of hearing or less. Out-of-scope chains include city programs, the city’s own transactions, rent or tenant matters, and paper land actions. Candidate conditions are rejected when they occur on an out-of-scope chain, do not represent an adopted condition, or bind the city or a third party rather than the applicant. The 12,193 verified conditions comprise 8,725 negotiated, 3,464 mandatory, and 4 unclear conditions. “Multi-meeting” means at least two agenda items; 37 of the 1,734 residential chains appear more than once within a single meeting.

7.2 Annotation of Project Conditions

Conditions of approval are extracted from the 5,802 project chains in Panel A: all 2,394 multi-item chains and the 3,408 single-item chains with more than five minutes of hearing time. For each chain, an LLM reads the full transcript record and the chain’s stage history and returns each condition with a category, a short description, and a mandatory-vs.-negotiated call. The conditions block of the extraction prompt is below (the same call also extracts project attributes and between-meeting changes; those blocks are omitted here).

Extraction Prompt (conditions block)

You are extracting land-use project information from a chain of city council meeting records about a single project. [...] Extract what the CONTEXT ACTUALLY SHOWS. If an attribute value is not stated, OMIT IT -- do not guess. Every evidence_quote must be a near-verbatim substring of the provided context.

CONDITIONS OF APPROVAL (per stage: the STOCK, not just deltas). For every stage, emit the full list of conditions present at that stage. If carried forward, emit again. Categorize each from this set: design_aesthetic, operational_hours, use_restriction, access_circulation, landscaping_buffer, affordability_covenant, tenant_protections, community_benefit, construction_management, environmental_monitoring, phasing_timeline, public_art, noise_lighting, tree_preservation, parking_management, traffic_mitigation, other.

For each condition fill: description (≤ 200 chars); binding: condition_of_approval | advisory | voluntary_commitment; imposed_by: council_motion | staff_report | settlement | state_mandate | applicant_proffer | planning_commission | design_review_board | local_ordinance (standard CUP / municipal-code recital) | prior_entitlement (carried forward from prior approval, no new negotiation); bucket: mandatory | negotiated | unclear (mandatory = required by statute, court, or state mandate, no real negotiation; negotiated = a discretionary ask added by council/staff/applicant; unclear = boilerplate

where the line is genuinely fuzzy); bucket_reasoning; dollar_amount_usd when a specific dollar value is tied to this condition; evidence_stage_idx where the condition was first imposed.

De-duplicating restatements of the same condition across a chain's hearings leaves 17,809 unique candidate conditions across 5,247 project chains. My main concern with this output was over-extraction: hand-reading early samples showed that alongside true conditions it captured items from chains that are not development projects at all (city programs, the city's own land deals, rent-board matters), project features and unadopted public demands described in the same breath as conditions, and obligations that bind the city rather than the applicant. I therefore ran a second verification pass over every extracted condition:

Verification Prompt

You audit "conditions of approval" that an earlier model extracted from California city-council land-use hearing transcripts. You receive one CHAIN (all hearings on one project/matter): its name(s), city, dates, chain-wide STAGE SUMMARIES, and a numbered list of extracted conditions, each with a transcript excerpt. Return STRICT JSON [...]

CHAIN SCOPE. in_scope=true only for private development / entitlement matters -- a private (or nonprofit/public-private) applicant seeking approval to build, expand, convert, subdivide-for-development, or entitle a specific project. Affordable-housing DEVELOPMENT projects are in scope. NOT in scope: city_program_or_agreement (city service programs, operating MOUs/licenses with nonprofits, e.g. a shelter OPERATIONS agreement); city_own_transaction (the city buying/selling/exchanging its own property); rent_or_tenant_matter; paper_land_action (lot mergers/subdivisions "for conveyance purposes only"); other_non_development.

cond_type (judge each extracted condition from the excerpt AND the stage summaries): condition = an obligation BINDING ON THIS PROJECT's approval -- imposed by council/staff/commission, adopted into a motion or development agreement, OR required by code/ordinance/CEQA and applied to this project. Adopted community-benefit or land-dedication packages count as condition (negotiated), even when the developer nominally "offered" them. applicant_offer = a voluntary proffer NOT (yet) adopted into the approval. project_feature = an attribute of the project as proposed; nothing imposed or offered in exchange for approval. When torn between applicant_offer and project_feature, choose project_feature. demand_only = someone asked for it and it was NOT adopted. Before choosing this, CHECK THE STAGE SUMMARIES for adoption at a later hearing; if adopted later, it is a condition. discussed_only = described in the abstract (what a zone permits generally, another project's terms, a contingent future process); no obligation attached. relief_granted = a waiver/variance/reduction granted TO the applicant. other.

IMPORTANT PRESUMPTION: your excerpt is a PARTIAL window; the extractor saw the full record. Hearings routinely adopt the staff-recommended conditions in one final motion that may fall outside your excerpt. If the excerpt shows the item being stated as a requirement/condition FOR THIS PROJECT, classify it as "condition" -- do NOT demote to discussed_only or demand_only merely because you cannot see the adoption vote in the excerpt.

bound_party: who bears the obligation -- "applicant" (developer/property owner), "city" (the city itself pays or acts), "third_party", or "unclear".

negotiated: "mandatory" = standard/code-required/CEQA-required, applies to every such project; "negotiated" = bargained, discretionary, or extracted/offered in exchange for approval; "unclear" if the record doesn't say.

Judge only from the provided text. Do not guess what the extractor "probably" meant.

A condition enters the final dataset if its chain is a private development matter and the item is a condition or adopted applicant offer binding the applicant. This retains 12,193 of the 17,809 candidates (68.5%) on 4,232 of the 5,247 chains, of which 8,725 are classified as negotiated and 3,464 as statutorily mandated.

Validation. I hand-labeled condition rows in three rounds, blind to the models' labels in each. The first two rounds (40 rows each, drawn from the raw extraction) are what motivated and shaped the verification pass—only 58% of the first sample were real conditions, and my margin notes on those rows became the verification prompt's scope and type rules (the second sample, under the revised protocol, came in at 80%). The reported validation is a third, held-out sample drawn after the verification pass: 40 fresh rows from chains I had not previously seen, half from conditions the verifier retained and half from conditions it dropped, with the strata hidden and rows in random order. I labeled each row from the condition text and a transcript window, revisiting disagreements against wider windows before finalizing.¹⁷

¹⁷Two protocol notes. First, the labeling form initially lacked an "out of scope" option, which forced a handful of city-program rows into condition-shaped answers; I re-ruled those once the option was added. Second, several disagreement rows were discussed with a higher reasoning model—Claude Fable 5—against the Gemini model's verdicts before my final pass; I ultimately sided against the model on about half of them, so the final labels are not simply anchored to the model.

Of the 20 retained conditions, I agreed with 19 (95%; 95% CI [76, 99]); the one exception was a row where the extractor had recorded a project’s \$45 million land purchase price as if it were a condition. Of the 20 dropped rows, I judged 3 (15%, [5, 36]) to be real conditions the verifier should have kept—all three are obligations whose adoption language is scattered across the record. The audit deliberately sampled equal numbers of retained and dropped rows, whereas the full candidate pool is 68.5% retained and 31.5% dropped. Reweighting the two audit strata to match the population implies that the verifier retained approximately 93% of the real conditions and agreed with my labels on approximately 92% of all candidate rows. In other words, the final dataset omits some genuine conditions, but contains relatively few apparent nonconditions.

Project size changes. The same extraction produces a candidate change log for quantitative project attributes, including unit count, building height, parking, and affordable units. For each purported change, the model records the before and after values. The raw log contains 752 candidate quantitative changes, of which 484 concern unit counts. A 39-row human calibration audit found that only about half represented genuine changes in a project’s total number of dwelling units. I therefore screen every unit-count entry before using it. This leaves 404 candidate increases or decreases. An LLM verification step then checks whether both numbers refer to the same project’s total dwelling-unit count and whether the transcript describes an actual revision; 256 claims, or 63 percent, pass this screen. I refer to these as screen-passing rather than human-verified changes. They occur in 241 projects and are nearly evenly divided between decreases and increases (140 versus 116). I recompute magnitudes from the retained before and after values: the median decrease is 12 percent and the median increase is 17 percent. The excluded claims often compare a proposal with zoning capacity, an alternative under discussion, or a single phase with the entire project, which helps explain why changes in the raw log appear larger.

Validation of the unit-count change log. To validate the screen, I hand-labeled a stratified sample of 39 raw entries, judging from the transcripts whether each logged change was a real revision of the project’s unit count. Only 51 percent [36, 66] of raw entries were—which is why the screen is needed. The screen’s keep-or-drop decisions agree with my labels 87 percent of the time [73, 94]: rows it keeps are genuine changes 85 percent of the time [64, 95], and it retains 89 percent [69, 97] of the genuine changes.

7.3 Participation and Project Conditions

Data. The unit of analysis is a development project followed across its city council appearances, documented in the data as land-use agenda items linked across meetings into 1,734 multi-meeting project chains.¹⁸ The verified set of 12,193 conditions is the outcome measure throughout this section. Participation is measured at the project’s first observed hearing with public comment, restricted to the public: nongovernmental speakers, excluding business- and applicant-affiliated ones.¹⁹ Comment stance comes from a per-comment attitude score described earlier (opposed below -0.5 , supportive above $+0.5$); sample projects draw a mean of 2.9 opposed and 2.1 supporting speakers at that hearing.

Level margin. The first question is whether participation predicts how many conditions attach. The estimation sample is the 611 in-scope project chains with at least one public comment at a first hearing, of which 491, in 91 cities and first heard 2015–2024, have complete covariates. For project j in city c first receiving comment in year t , I estimate by OLS:

$$\log(1 + C_j) = \beta_S \log(1 + S_j) + \beta_O \log(1 + O_j) + X_j' \gamma + \delta_c + \tau_t + \varepsilon_j, \quad (3)$$

where C_j counts the project’s verified conditions and S_j and O_j count supporting and opposed speakers at the first hearing with comment. X_j contains log proposed units, an affordability indicator and the affordable share, and project-type indicators; log tract median household income, log tract median home value, and housing density; and two record-length controls—log agenda items and log meetings in the chain—so longer project records cannot mechanically carry both more recorded comment and more extracted conditions. δ_c and τ_t are city (91) and first-comment-year (10) fixed effects; standard errors are clustered by city.

92 percent of sample projects carry at least one verified condition, with a mean of 4.2, leaving little extensive-margin variance for participation to explain. Panel A of Table 21 reports the estimates: the supporting- and -opposed-speaker coefficients are small, statistically indistinguishable from zero, and indistinguishable from each other in every column. Splitting conditions into negotiated terms and mandatory code requirements changes nothing, and extensive-margin linear probability models are equally flat.

¹⁸I first attempted condition extraction in the San Francisco Planning Commission records I used for the delay. Conditions in these records are observable only in the staff packet as a standardized “Exhibit A.” However, these were almost entirely boilerplate recitals that seemed to bear no relation to commenters, and the minutes proved too sparse to tie conditions to comments. The council transcripts, on the other hand, capture conditions as negotiated in the room and the full text of every comment and add cross-city variation.

¹⁹The restriction matters: under the raw “public comment” flag, half of supportive speech at these hearings comes from council members and staff, and a further third of the nongovernmental remainder from the applicant’s own team.

Composition margin. What participation does predict is *which* conditions attach. I classify each of the 3,491 public comments at first-comment hearings, multi-label, into the sixteen substantive categories used for conditions (the extraction prompt’s seventeenth label, the residual *other*, is excluded on both the comment and condition sides of the panel), and estimate, for project j and condition category c ,

$$\text{Condition}_{jc} = \beta \text{Raised}_{jc} + \alpha_j + \gamma_c + \varepsilon_{jc}, \quad (4)$$

where $\text{Condition}_{jc} = 1$ if project j ’s record contains a verified condition of category c , and $\text{Raised}_{jc} = 1$ if a member of the public raised category c at project j ’s first hearing with comment. Each project contributes sixteen observations, one per condition category, for a panel of $611 \times 16 = 9,776$ cells. The project effects α_j give each project its own intercept, so anything constant about the project (city, size, year, how contested it was, how many conditions it drew in total, how many people spoke) cannot move the estimate, which is why the specification carries no additional controls. The category effects γ_c give each category its own baseline, so the estimate is not inflated by categories that are common on both sides of the equation everywhere. β is identified by the comparison, within a single project, of the categories the public raised against the categories it did not. Standard errors are clustered by city (96).

Panel B of Table 21 reports the estimates. A category raised at the first hearing is 9.4 percentage points more likely to appear among the project’s verified conditions than a category no commenter raised, against a base rate of 15.9 percent—roughly 60 percent above baseline. The estimate is robust to replacing γ_c with city $\times$ category fixed effects and to restricting the panel to the level-regression sample (+9.5 percentage points, SE 1.4), and it is positive and significant for 11 of 16 categories individually.

Two further checks establish what this correlation is, and what it is not. First, it is not an artifact of the condition-verification pass. One might worry that the pattern is mechanical—that the extraction or verification step is more likely to record a condition when nearby comment text mentions the same subject. But the same regression run on the raw extracted conditions, before any verification, yields a proportionally similar estimate (+10.5 points on an 18.7 percent base), so verification did not create the pattern; and run on the candidate conditions the verification *rejected*—where extraction noise concentrates—the relationship is proportionally weaker (+2.5 points on a 4.9 percent base). Second, the match is category-specific. Comparing raw means category by category, conditions of a category are on average 1.72 times as likely to appear when commenters raise that category as the category’s unconditional base rate, while conditions in *other* categories barely move (mean lift 1.02).²⁰ Projects that draw comment do not simply accumulate conditions of every kind; the condition that appears is the one on the subject commenters raised.

²⁰Both lifts average cells of the 16×16 matrix of raised-category by condition-category raw conditional means over unconditional category means—the diagonal (1.72) and the off-diagonal (1.02). Because this comparison equal-weights the sixteen categories and adjusts for nothing, the diagonal average is larger than the pooled lift implied by the regression estimate, $(15.9 + 9.4)/15.9 \approx 1.6$.

Table 21: First-hearing participation and approval conditions: level and composition

	(1) All conditions	(2) Negotiated only	(3) Mandatory only
<i>Panel A. Level: project level, log(1 + conditions)</i>			
log(1 + supporting speakers)	0.017 (0.048)	0.017 (0.055)	−0.015 (0.040)
log(1 + opposed speakers)	0.017 (0.046)	0.017 (0.041)	0.015 (0.050)
Mean of dependent variable	1.42	1.18	0.53
Observations (projects)	491	491	491
R^2	0.409	0.414	0.298
<i>Panel B. Composition: project × category level, any condition in category</i>			
Category raised at first hearing	0.094*** (0.012)	0.067*** (0.010)	0.034*** (0.007)
... raised in opposition	0.069*** (0.015)	0.054*** (0.014)	0.023** (0.009)
... raised in support	0.086*** (0.017)	0.064*** (0.014)	0.026** (0.011)
Mean of dependent variable	0.159	0.121	0.048
Observations (project × category)	9,776	9,776	9,776
R^2	0.187	0.173	0.129

Notes: Column (1) uses all verified conditions, and columns (2)–(3) split by negotiated/mandatory label. Panel A estimates equation (3) by OLS on the 491 in-scope project chains with a public comment at the first hearing with comment and complete covariates; the dependent variable is log(1 + conditions) anywhere in the project's record, and project, record-length, and tract controls plus city (91) and first-comment-year (10) fixed effects are absorbed, not printed. Extensive-margin LPMs are equally flat: any negotiated (base 0.841), supporting 0.030 (0.025), opposed 0.012 (0.024); any mandatory (base 0.544), supporting 0.018 (0.036), opposed −0.015 (0.048); any condition (base 0.923), supporting 0.036 (0.019), opposed −0.000 (0.021)—the largest of these, supportive speakers on the probability of any condition, is significant only at the ten-percent level. Panel B estimates equation (4) as a linear probability model on the 611 × 16 project–category panel with project and condition-category fixed effects; the dependent variable equals one if the project's record contains a verified condition of the category, and *category raised* equals one if a public commenter raised it at the first hearing with comment. The top row enters the pooled indicator; the opposition/support rows are a separate specification entering both jointly; their equality is never rejected ($p \geq 0.50$). Standard errors clustered by city in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At least two readings of this pattern are possible. The first is responsiveness: the council hears the concern and writes the condition. In Menifee, for example, neighbors on one-acre horse properties objected to two-story homes overlooking their lots, and the approval added an eight-foot wall and a one-story limit on the facing street. The second is shared observation: a single site attribute can prompt both the complaint and the condition with no transmission between them—the adjacency that brought the Menifee neighbors to the microphone is the same adjacency a planner reviewing the site plan might have conditioned on regardless, and because such an attribute varies across categories *within* a project, the fixed effects in equation (4) do not rule it out. To gauge the prevalence of the second reading, I audit two hundred matched cases—a condition in a category the public raised, recorded at the hearing where they raised it—and use the diarized, timestamped transcript to locate where in the hearing the condition's substance first appears.²¹ The result is an almost even split. In half (45.7 percent), the condition appears first in the staff presentation, before any member of the public has spoken. In the other half (45.7 percent), the condition is first articulated only *after* the public comments—in a council member's motion, a staff revision, or a concession the applicant offers in the room (6.7 percent are not locatable in the excerpt). Only two percent (1.8) of audited conditions exist solely in public comment, consistent with the verification pass, which independently types just 2.8 percent of extracted candidates as unadopted podium demands and, in the verified set used here, removes them by construction. Either reading supports the conclusion that the public hearing operates as an exchange over a project's terms.

²¹The audit stays within the hearing deliberately: most condition packages originate below the council, in staff review and at the planning commission, which these transcripts do not observe, so timing *across* hearings cannot separate the two readings. Each audit verdict carries a timestamped near-verbatim quote.

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